

On Functional GARCH Models and Their Application to Intraday Value-at-Risk

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1 Introduction

Risk management aims to identify, measure, and evaluate potential losses in order to support decision-making under uncertainty. To this end, risk managers employ specific risk measures to assess the exposure associated with a particular decision or investment. One of the most widely used measures of market risk, defined as the risk of losses arising from on- and off-balance-sheet positions, is value-at-risk (VaR). Intuitively, VaR is intended to specify a threshold below which returns fall only with a predetermined probability. VaR was incorporated into financial regulation through the 1996 Market Risk Amendment to the Basel I Accord, which allowed banks to use internal VaR models for the calculation of regulatory capital and remains a widely used and important regulatory risk measure til today. Since forecasting of VaR requires an assessment of the conditional distribution of returns, models for time-varying volatility play a central role in its practical implementation. In particular, one common approach to forecasting VaR is to model the conditional volatility of financial returns using (generalized) autoregressive conditional heteroskedasticity, or (G)ARCH, models. These models differ fundamentally from standard time series models such as autoregressive moving average (ARMA) or vector autoregressive (VAR) models. Rather than focusing on the conditional mean, (G)ARCH models describe the conditional variance by relating current volatility to past squared returns and past volatility. By explicitly modelling conditional volatility, these type of models provide a natural framework for time-varying financial risk and thus for the forecasting of VaR.

The increasing availability of high-frequency financial data, together with the growth of high-frequency trading (Menkveld, 2016; Zaharudin et al., 2022; Huang et al., 2025; Staff of the Market Intelligence Branch, Division of Market Oversight, 2019; Bouveret et al., 2014) and the demand for timely risk assessment, has intensified the need for methods that capture intraday volatility dynamics. Given the widespread use of GARCH models for forecasting daily VaR, a natural question is whether such models can be extended to describe intraday volatility and used to forecast intraday VaR. This thesis addresses these questions by studying the Functional GARCH (FGARCH) model, which extends the classical GARCH framework to a functional setting and thereby provides a flexible approach to modelling intraday volatility curves.

This thesis combines a review of the theoretical foundations of the FGARCH model with

an original simulation-based analysis of its finite-sample behaviour and forecasting performance. On the theoretical side, it summarizes key results from the literature concerning the existence of strictly stationary and non-anticipative solutions, the existence of moments, ergodicity, weak dependence, and the large-sample properties of two established estimation procedures for the functional parameters. On the empirical side, the Monte Carlo study shows that, in a simple simulation setting in which the true underlying basis function is assumed to be known and only one basis function is used, (pseudo) quasi-maximum likelihood (QML) estimation outperforms least-squares (LS) estimation. The results further indicate that thinner-tailed innovation increments and larger sample sizes lead to more accurate parameter estimates and improved intraday VaR forecasts. When the true underlying basis functions are not known and the assumptions underlying the estimation procedures are violated, the comparison becomes less clear. In particular, both methods perform similarly when the assumption that all functional parameters can be represented using a common set of basis functions is violated. Finally, the results show that the PC-GARCH model performs relatively well compared with the FGARCH model even when the data are generated from an FGARCH process.

When discretized intraday observations are viewed as realizations of curves over a trading day, methods from Functional Data Analysis (FDA) offer a natural framework for their analysis. FDA studies data that are represented as functions or curves; see, for example, Ramsay and Silverman (2005), Hsing and Eubank (2015), and Kokoszka and Reimherr (2017). More recently, ideas from volatility modelling and functional time series analysis have been combined in order to study financial data at the intraday level. Hörmann et al. (2013) extends the ARCH framework to the functional setting, and Aue et al. (2017) subsequently develops a functional version of the GARCH model. Building on this framework, Cerovecki et al. (2019) proposes a QML estimation approach intended to improve on the LS procedure considered in Aue et al. (2017). In a related application, Rice et al. (2020a) uses functional GARCH-type models to forecast intraday VaR curves. This thesis builds on this literature. However, the available results are technically demanding and often presuppose substantial background knowledge in functional data analysis and financial econometrics. As a result, the existing literature may be difficult to navigate for readers seeking a structured introduction to the model's theoretical background, estimation, and practical use. The present thesis therefore aims to summarize and contextualize the existing literature while also examining aspects that are particularly relevant for applications.

In this sense, the thesis is intended both as an introduction to the FGARCH framework and as a simulation-based investigation of its use for intraday VaR forecasting.

This thesis proceeds as follows. Section 2 presents the main theoretical results on the FGARCH model and introduces VaR. Section 3 then examines the finite-sample performance of the FGARCH model and its use for forecasting intraday VaR by means of Monte Carlo simulations. Finally, Section 4 concludes and discusses possible extensions of the FGARCH model as well as directions for future research.

2 Functional GARCH

This section introduces the FGARCH model, discusses its theoretical properties, and presents methods for estimating its functional parameters. In addition, one subsection is devoted to the introduction of the concept of VaR. The section is structured as follows. First, Section 2.1 provides a selective overview of the main models and results for univariate and multivariate GARCH processes. Second, Section 2.2 presents the FGARCH model and discusses its theoretical properties. Third, Section 2.3 presents two estimation methods for the FGARCH model and reviews their consistency and asymptotic distributions. Fourth, Section 2.4 introduces a mathematical definition of VaR, discusses its history and main theoretical properties, and reviews important criticisms of the measure.

2.1 Univariate GARCH and Multivariate GARCH

Before introducing the functional GARCH model, I first review the univariate GARCH model and then its multivariate extension. I begin with the univariate case, since it provides the basic template for the extensions considered later. The GARCH model, introduced by Bollerslev (1986) is a generalization of the ARCH model of Engle (1982). A univariate GARCH(p, q) model can be described by its first two moments:

Definition 2.1 (GARCH(p, q) Process). A process (y_t) is called a GARCH(p, q) process if its first two conditional moments exist and satisfy:

1. $E(y_t | y_u, u < t) = 0, \quad \forall t \in \mathbb{Z}.$

2. There exist constants $\delta > 0$, $\alpha_i \geq 0$, $i = 1, \dots, q$ and $\beta_j \geq 0$, $j = 1, \dots, p$ such that

$$\sigma_t^2 = \text{Var}(y_t | y_u, u < t) = \delta + \sum_{i=1}^q \alpha_i y_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad (1)$$

Thus, GARCH models specify the current conditional variance as a function of past squared returns and past conditional variances. This approach is motivated by well-known stylized facts of financial time series: while returns typically display little autocorrelation, squared or absolute returns often exhibit strong and persistent dependence, and volatility tends to cluster over time (Mandelbrot et al., 1963; Taylor, 2011).

To relate the GARCH(p, q) model to more common linear time series structures, define the innovation term $\nu_t = y_t^2 - \sigma_t^2$ and substitute $\sigma_t^2 = y_t^2 - \nu_t$ into the volatility equation

$$y_t^2 = \delta + \sum_{i=1}^r \overbrace{(\alpha_i + \beta_i)}^{\kappa_i} y_{t-i}^2 + \nu_t - \sum_{j=1}^p \beta_j \nu_{t-j} = \delta + \sum_{i=1}^r \kappa_i y_{t-i}^2 + \nu_t - \sum_{j=1}^p \beta_j \nu_{t-j} \quad (2)$$

where $r = \max(p, q)$ with $\alpha_i = 0$ ($\beta_j = 0$) if $i > q$ ($j > p$). Hence, equation (2) shows that the squared process (y_t^2) admits an ARMA(r, p)-type representation. This also clarifies in what sense the GARCH model extends the ARCH model: just as an ARMA process extends an AR process by including lagged moving-average terms, the GARCH model extends the ARCH model by allowing lagged conditional variances to enter the volatility recursion. Moreover, the analogy goes one step further. Just as an ARMA process can, under suitable conditions, be represented as an AR(∞) process, a GARCH(p, q) process can likewise be represented as an ARCH(∞) process (Francq and Zakoian, 2019). It should be noted, however, that $(\nu_t)_{t \in \mathbb{Z}}$ is generally not i.i.d. Therefore, the representation in equation (2) should not be treated as a standard ARMA model when studying the existence of stationary solutions to a GARCH(p, q) process.

A more restrictive, but often more convenient, formulation is the strong GARCH(p, q) process, which specifies the model through an explicit innovation process. A strong univariate GARCH(p, q) process is given by:

$$y_t = \sigma_t \varepsilon_t \quad (3)$$

$$\sigma_t^2 = \delta + \sum_{i=1}^q \alpha_i y_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2 \quad (4)$$

where $(\varepsilon_t)_{t \in \mathbb{Z}}$ is i.i.d. This specification implies the preceding moment conditions and is therefore a stronger formulation of the GARCH model. In the following, the focus will be on the strong GARCH(1, 1) process, since the corresponding functional version is studied in Aue et al. (2017) and applied in Rice et al. (2020a). The strong GARCH(1, 1) process is the simplest specification within the class of strong GARCH models. Despite its simplicity, it is also one of the most widely used specifications in practice (Zivot, 2009).

To study the existence of a stationary solution to the strong GARCH(1, 1) process, it is useful to rewrite the volatility equation as

$$\sigma_t^2 = \delta + \gamma_{t-1} \sigma_{t-1}^2, \quad \text{where } \gamma_t = \alpha \varepsilon_t^2 + \beta. \quad (5)$$

This representation shows that the volatility equation of the strong GARCH(1, 1) process can be viewed as an autoregressive model with random coefficients. This representation is useful because stationarity can be studied by imposing conditions on $(\gamma_t)_{t \in \mathbb{Z}}$ alone, rather than separately on α , β , and $(\varepsilon_t)_{t \in \mathbb{Z}}$.

The following condition taken from Francq and Zakoian (2019) guarantees the existence of a strictly stationary, non-anticipative, and ergodic solution to the strong GARCH(1, 1) process:

Theorem 2.1 (Strict stationarity of the strong GARCH(1, 1) process). If

$$-\infty \leq \mathbb{E}[\log\{\overbrace{\alpha \varepsilon_t^2 + \beta}^{\gamma_t}\}] < 0, \quad (6)$$

then the infinite sum

$$h_t = \delta \left(1 + \sum_{i=1}^{\infty} \Gamma_{t,i} \right), \quad \text{where } \Gamma_{t,i} = \gamma_{t-1} \gamma_{t-2} \cdots \gamma_{t-i} \quad (7)$$

converges almost surely, and the process (y_t) defined by $y_t = \sqrt{h_t} \varepsilon_t$ is the unique strictly stationary solution of the strong GARCH(1, 1) model. This solution is non-anticipative and ergodic.

To prove Theorem 2.1, note that Equation (4) can be solved recursively to obtain h_t . Showing that h_t converges involves an application of the Cauchy root test as well as the strong law of large numbers. Uniqueness is then established by contradiction: assuming

the existence of another solution, one shows that it must coincide almost surely with the original one.

Ergodicity describes the property of a stochastic process that time averages along a single, sufficiently long realization converge to the same values as ensemble averages taken over many realizations. A particularly strong sufficient condition for ergodicity is geometric ergodicity, which requires not only convergence to a unique stationary distribution, but convergence at an exponential rate. It can be shown that the conditional volatility of the GARCH(1, 1) process $(\sigma_t; t \in \mathbb{Z})$ is geometrically ergodic under suitable regularity conditions. Intuitively, this means that the effect of initial conditions vanishes exponentially fast and that sample averages converge to their population counterparts. Moreover, $(y_t; t \in \mathbb{Z})$ is geometrically β -mixing, a strong dependence property that implies ergodicity. To prove that the GARCH(1, 1) process is geometrically ergodic first note that the volatility equation of the GARCH process can be rewritten into a Markov Chain representation. One can then make use of a Theorem from Feigin and Tweedie (1985). They show that a Markov chain that is also an aperiodic, irreducible Feller chain is geometrically ergodic if it satisfies a drift condition. Proving that $(y_t; t \in \mathbb{Z})$ is geometrically β -mixing follows from the geometric ergodicity of $(\sigma_t; t \in \mathbb{Z})$ and the fact that $(\varepsilon_t; t \in \mathbb{Z})$ is i.i.d. $(y_t; t \in \mathbb{Z})$ is not geometrically ergodic since it is not Markovian. The complete proof can be found in Francq and Zakoian (2019). A precise mathematical definition of ergodicity is given in Section A.2.1.

Remark 2.1: Physical intuition behind mixing and ergodicity

A useful intuition for statistical mixing and ergodicity can be obtained from the physical act of stirring different liquids in a glass. Imagine that the glass is divided into many tiny regions and that we observe which liquid occupies each region after every stir. If we record the configuration after each stir, the resulting sequence of observations can be interpreted as a realization of a stochastic process, while each stir corresponds to one iteration of its transition mechanism.

Ergodicity intuitively means that, after many stirring steps, the liquids become distributed throughout the glass according to their overall proportions. In other words, the long-run fraction of time that a given liquid is observed in a particular region corresponds to the amount of that liquid in the glass and the size of the region. Geometric ergodicity strengthens this idea by stating that the distribution of liquids across the

glass approaches this long-run configuration at an exponential rate.

The notion of mixing describes how quickly the influence of the past on the future disappears. Mixing coefficients such as the α - and β -mixing coefficients quantify how close observations that are far apart in time are to being statistically independent. In the stirring analogy, after sufficiently many stirs the spatial arrangement of liquids becomes nearly independent of how the liquids were initially poured into the glass. Roughly speaking, the α -mixing coefficient measures the remaining dependence between past and future events, while the β -mixing coefficient provides a stronger measure of how far their joint behavior is from statistical independence. Both therefore quantify how effectively the system "forgets" its earlier states.

This intuition also illustrates the relationship between ergodicity and mixing. If the liquids mix very efficiently—like highly fluid liquids—then the distribution in the glass quickly approaches its long-run configuration and the dependence between distant observations vanishes rapidly. This corresponds to geometric ergodicity and typically implies strong mixing properties. By contrast, if the liquids are more viscous, the system may still eventually explore all regions of the glass (ergodicity), but the approach to the long-run distribution and the loss of dependence can occur much more slowly.

Next, I will provide a short exposition of multivariate GARCH processes, as they are very closely related to FGARCH processes in practice. Defining a multivariate GARCH process is not as straightforward as in the univariate case. This is because moving from the univariate to the multivariate setting requires not only the modeling of additional conditional variances, but also the conditional covariances between the components. Without further restrictions, the resulting number of parameters quickly becomes too large for reliable estimation and asymptotic analysis. Therefore, the conditional covariance matrix H_t follows an explicit specification. In the following, I will present two of these specifications since they are closely connected to the FGARCH model. Throughout this discussion, I will assume that $y_t, \varepsilon_t \in \mathbb{R}^m$ and that ε_t is an i.i.d. vector.

First, I consider the PC-GARCH model, which is defined by

$$y_t = H_t^{\frac{1}{2}} \varepsilon_t, \tag{8}$$

$$H_t = P \Lambda_t P^\top. \tag{9}$$

Here, P is an $m \times m$ orthogonal matrix, while $\Lambda_t = \text{diag}(\lambda_{1,t}, \lambda_{2,t}, \dots, \lambda_{m,t})$ is an $m \times m$ diagonal matrix. The matrix P is typically estimated using Principal Component Analysis, and the diagonal elements of Λ_t are assumed to follow univariate GARCH(1,1) processes:

$$\lambda_{i,t} = \delta_i + \alpha_i f_{i,t-1}^2 + \beta_i \lambda_{i,t-1}, \quad i = 1, \dots, m. \quad (10)$$

In this specification, $f_{i,t}$ denotes the i -th component of the transformed series $f_t = P^\top y_t$. The parameters in equation (10) are typically estimated by QML. From the representation in equation (9), it follows directly that the PC-GARCH model decomposes the conditional covariance matrix H_t into a set of constant eigenvectors and time-varying eigenvalues.

The second multivariate GARCH specification considered is the Constant Conditional Correlation (CCC) GARCH model, which is defined as

$$y_t = H_t^{\frac{1}{2}} \varepsilon_t, \quad (11)$$

$$H_t = D_t R D_t^\top. \quad (12)$$

Here, $D_t = \text{diag}(\sqrt{h_{1,t}}, \sqrt{h_{2,t}}, \dots, \sqrt{h_{m,t}})$ is an $m \times m$ diagonal matrix and R is an $m \times m$ constant correlation matrix. The vector of conditional variances h_t is assumed to follow a GARCH(1,1)-type specification given by

$$h_t = \delta + A(y_{t-1} \odot y_{t-1}) + B h_{t-1}, \quad (13)$$

where $\odot$ denotes the elementwise (Hadamard) product, δ is an m -dimensional vector, and A and B are $m \times m$ parameter matrices. Equation (12) shows that H_t consists of time-varying, variable-specific conditional variances combined with a constant correlation structure. In addition, equation (13) shows that the variable-specific conditional variances evolve dynamically, depending not only on each variable's own past but also on the past of the other variables.

Since in practice only the process $(y_t)_{t \in \mathbb{Z}}$ is observed, the parameters of any univariate or multivariate GARCH model have to be estimated. GARCH models are typically estimated using QML estimation. For GARCH models, QML estimation is generally preferred to LS estimation, since LS methods require stronger moment conditions and are typically

less efficient (Francq and Zakoian, 2019). In the following, I will illustrate the estimation procedure for the univariate GARCH(1,1) model. Analogous arguments apply to other GARCH specifications.

The Gaussian quasi-likelihood for the GARCH(1,1) model is given by

$$L_n(\underline{\theta}; y_1, \dots, y_n) = \prod_{t=1}^n \frac{1}{\sqrt{2\pi\tilde{\sigma}_t^2}} \exp\left(-\frac{y_t^2}{2\tilde{\sigma}_t^2}\right); \quad \tilde{\sigma}_t^2 = \tilde{\sigma}_t^2(\underline{\theta}) = \underline{\delta} + \underline{\alpha}y_{t-1}^2 + \underline{\beta}\tilde{\sigma}_{t-1}^2. \quad (14)$$

Here, $\underline{\theta} = (\underline{\delta}, \underline{\alpha}, \underline{\beta})^\top$ denotes the parameter vector, and $\tilde{\sigma}_t^2$ is computed recursively. The true parameter vector is denoted by $\theta = (\delta, \alpha, \beta)^\top$.

Taking logarithms, omitting constants and multiplying by -1 , the optimization problem is equivalent to minimizing

$$\mathcal{I}_n = \frac{1}{n} \sum_{t=1}^n \left(\frac{y_t^2}{\tilde{\sigma}_t^2} + \log \tilde{\sigma}_t^2 \right). \quad (15)$$

The parameter estimator is then obtained numerically by minimizing this criterion, for example by using a Newton-Raphson algorithm.

Under suitable regularity conditions, the QML estimator is consistent and asymptotically normal. If the innovations are Gaussian, its asymptotic variance is given by the inverse Fisher information matrix. If the innovations are non-Gaussian, the asymptotic variance instead involves the inverse of an adjusted information matrix. The proof can be found in Francq and Zakoian (2019).

2.2 Model

This section introduces the FGARCH model and discusses its key theoretical properties. I begin with the definition of the model.

A FGARCH process is defined as follows:

Definition 2.2 (FGARCH(p,q) Process). A sequence of random functions $(y_i(t))_{i \in \mathbb{Z}}$ is

called a FGARCH process of order (p, q) if it satisfies the following equations

$$y_i(t) = \sigma_i(t)\varepsilon_i(t) \quad (16)$$

$$\sigma_i^2(t) = \delta(t) + \sum_{j=1}^q \alpha_j y_{i-j}^2(t) + \sum_{k=1}^p \beta_k \sigma_{i-k}^2(t). \quad (17)$$

Here, $\delta(t) \in L^2[0, 1]$ is a non-negative function, and $(\alpha_j)_{j \in (1, \dots, p)}$ and $(\beta_k)_{k \in (1, \dots, q)}$ are bounded integral operators in $L^2[0, 1]^2$ given by $(\alpha_j x)(t) = \int \alpha_j(t, s)x(s)ds$ and $(\beta_k x)(t) = \int \beta_k(t, s)x(s)ds$ that map non-negative functions to non-negative functions for $x \in L^2[0, 1]$. By convention, I restrict the intraday time variable to $t \in [0, 1]$. This is without loss of generality since data that do not follow this convention can be converted into this format. Accordingly, $\int_0^1$ will be abbreviated by $\int$. The second time variable i denotes the trading day. The innovations $(\varepsilon_i(t))_{i \in \mathbb{Z}}$ are i.i.d. random functions.

From Definition 2.2, it is clear that the FGARCH(p, q) process is the functional analogue of the strong (scalar) GARCH process. As a consequence, the model admits an ARMA-type representation, just as in the univariate case discussed in equation (2).

Following Aue et al. (2017) and Rice et al. (2020a), I focus on the FGARCH(1, 1) specification, which is the functional counterpart of the benchmark strong (scalar) GARCH(1, 1) model. As discussed previously, despite its simplicity the strong GARCH(1, 1) process is also one of the most widely used specifications in practice. For this reason, the FGARCH(1, 1) model is a natural starting point in the functional setting. In this case, the FGARCH(p, q) equations reduce to

$$y_i(t) = \sigma_i(t)\varepsilon_i(t) \quad (18)$$

$$\sigma_i^2(t) = \delta(t) + \alpha y_{i-1}^2(t) + \beta \sigma_{i-1}^2(t). \quad (19)$$

In the following, conditions for the existence of a stationary solution to the FGARCH(1, 1) process are discussed. Aue et al. (2017) derives these results both for the space $L^2[0, 1]$, the space of square integrable functions on $[0, 1]$, as well as for the space of continuous functions $C[0, 1]$. I restrict attention to the $L^2[0, 1]$ setting, since the corresponding results for $C[0, 1]$ are of essentially the same form. The main difference is that the innovation functions and the intercept function $\delta(t)$ are assumed to belong to $C[0, 1]$, whereas the kernel of the integral operators remain in $L^2[0, 1]^2$, so that the resulting operators map

$C[0, 1]$ into $C[0, 1]$.

Using notation analogous to the scalar case, the volatility equation can be written as

$$\sigma_i^2(t) = \delta(t) + \gamma_{i-1}\sigma_{i-1}^2(t), \quad (20)$$

where γ_i is the random integral operator with kernel

$$\gamma_i(t, s) = \alpha(t, s)\varepsilon_i^2(s) + \beta(t, s). \quad (21)$$

In the following it is assumed that the kernels of α and β belong to $L^2([0, 1]^2)$ and that $\varepsilon_0 \in L^4[0, 1]$ almost surely. These assumptions imply that $\gamma_i \in L^2([0, 1]^2)$ with probability 1. Consequently, γ_i defines a bounded random integral operator on $L^2[0, 1]$; see Riesz and Nagy (1990). Similarly to the univariate case, writing the volatility recursion in this form is convenient, since the existence of a strictly stationary solution can then be characterized through properties of the random operator γ_i and, in particular, its Hilbert–Schmidt norm. The following theorem provides sufficient conditions for the existence of a strictly stationary solution.

Theorem 2.2 (Strict stationarity of the FGARCH(1, 1) process). Let $\delta(t) \in L^2[0, 1]$ be a non-negative function, and let α and β be integral operators with kernels in $L^2[0, 1]^2$ that map non-negative functions to non-negative functions. Let $\{\varepsilon_i(t)\}_{i \in \mathbb{Z}}$ be i.i.d. random functions where $\varepsilon_0 \in L^4[0, 1]$ almost surely.

(i) If

$$-\infty \leq \mathbb{E} [\log \|\gamma_0\|_{\mathcal{S}}] < 0 \quad (22)$$

then the infinite sum

$$h_i(t) = \delta(t) + \sum_{k=1}^{\infty} \Gamma_{i,k} \delta(t) \quad (23)$$

converges almost surely, and the process $\{y_i(t)\}_{i \in \mathbb{Z}}$, defined by $y_i(t) = \sqrt{h_i(t)}\varepsilon_i(t)$, is the unique strictly stationary solution to the FGARCH(1, 1) equations in (18) and (19). This solution is non-anticipative and ergodic.

(ii) If there is $\nu > 0$ such that

$$\mathbb{E} [\|\gamma_0\|_{\mathcal{S}}^\nu] < 1 \quad (24)$$

then there exists a unique, strictly stationary, and non-anticipative solution in $L^2[0, 1]$ to the FGARCH(1, 1) model and $\mathbb{E} [\|\sigma_0^2\|_2^\nu] < \infty$.

Here $\|\cdot\|_{\mathcal{S}}$ is the Hilbert-Schmidt norm defined as

$$\|\gamma_0\|_{\mathcal{S}} = \left(\int \int \gamma_0^2(t, s) dt ds \right)^{1/2} \quad (25)$$

and $\Gamma_{i,k}$ denotes the random integral operator

$$(\Gamma_{i,k}x)(t) = \int \cdots \int \gamma_{i-1}(t, s_1) \gamma_{i-2}(s_1, s_2) \cdots \gamma_{i-k}(s_{k-1}, s_k) x(s_k) ds_k \cdots ds_2 ds_1. \quad (26)$$

Due to the close resemblance between Theorem 2.2 part (i) and Theorem 2.1, the proof of (i) follows a very similar line of argument. First, $h_i(t)$ is obtained by recursively iterating equation (20). Second, almost sure convergence of $h_i(t)$ is established by combining the Cauchy–Schwarz inequality with the strong law of large numbers applied to the logarithms of the operator norms. Finally, the uniqueness of the solution is shown by exploiting the recursive structure of equation (20). Assuming the existence of an alternative solution implies that both solutions converge to the same limit and must therefore coincide, yielding a contradiction. The proof of Theorem 2.2 part (ii) proceeds by first establishing inequalities between $\|\sigma_0^2\|_2$ and the series $\sum_{k=0}^{\infty} \|\Gamma_{i,k}\delta\|_2$ using the Cauchy–Schwarz inequality. Applying Minkowski’s inequality then yields the claimed result on the existence of moments in (ii). The full proof of Theorem 2.2 can be found in Section A.3.1.

Both Cerovecki et al. (2019) and Kühnert (2020) extend the stationarity result in (i). Cerovecki et al. (2019) derives sufficient conditions for the existence of a unique strictly stationary and non-anticipative FGARCH(p, q) solution in L^2 for arbitrary orders p and q . Their approach is based on the top Lyapunov exponent (Kingman, 1973). They also note that, in contrast to the finite-dimensional case, establishing necessity is substantially more difficult in the functional setting because norms are not equivalent in infinite-dimensional spaces. When specialised to the FGARCH(1, 1) specification, their general result yields a milder sufficient condition than the condition in equation (22), although this comes at the

cost of imposing a different assumption on the innovation process. Kühnert (2020) further generalizes both the stationarity results and the result on the existence of moments in (ii) to L^p spaces with $p \in [1, \infty)$, and more generally to separable Banach spaces of functions on $[0, 1]$ endowed with the supremum norm and closed under pointwise multiplication.

To establish the ergodicity of the stationary solution, the following corollary is useful.

Corollary 2.1. The stationary non-anticipative solution established in Theorem 2.2 yields the following corollary:

- (i) There is a functional g such that

$$\sigma_i^2 = g(\varepsilon_{i-1}, \varepsilon_{i-2}, \dots). \quad (27)$$

As a consequence, $(\sigma_i^2(t); i \in \mathbb{Z})$ and $(y_i(t); i \in \mathbb{Z})$ are Bernoulli shifts. This follows directly from equation (23).

- (ii) Part (i) implies that both $\{\sigma_i^2(t)\}_{i \in \mathbb{Z}}$ and $\{y_i(t)\}_{i \in \mathbb{Z}}$ are ergodic.

Ergodicity of Bernoulli shifts is a well established result and follows from Stout (1974). In the functional setting, arguments of the type used to establish geometric ergodicity and β -mixing in the univariate case become much harder to apply. This is, for example, due to the fact that infinite-dimensional observations do not admit densities in the same way as finite-dimensional random variables. In addition, stronger assumptions on the functional parameters α and β may be needed to establish continuity properties of the functional recursion, similar to the role played by continuity in the proof of geometric ergodicity for the scalar GARCH model. Furthermore, showing that the FGARCH process forms an irreducible Markov chain is difficult in the infinite-dimensional setting, and establishing small sets is equally complicated. However, ergodicity alone is not sufficient for the derivation of important statistical results like central limit theorems. That is because information on how fast dependence decays is needed to establish convergence in distribution of suitable sample statistics. Since ergodicity is not sufficient for the application of the functional central limit theorem, Aue et al. (2017) derives the following result.

Corollary 2.2. (I) For $\ell \in \mathbb{N}$ and $j, i \in \mathbb{Z}$, let $\varepsilon_{i,j}^{(\ell)}(t)$ be independent copies of $\varepsilon_0(t)$

and define

$$\sigma_{i,\ell}^2(t) = g\left(\varepsilon_i(t), \dots, \varepsilon_{i-\ell}(t), \varepsilon_{i-\ell-1,i-\ell}^{(i)}(t), \varepsilon_{i-\ell-2,i-\ell}^{(i)}(t), \dots\right). \quad (28)$$

If equation (24) is satisfied, then there exists a constant $0 < \rho < 1$ such that

$$\mathbb{E} \left[\|\sigma_i^2 - \sigma_{i,\ell}^2\|_2^\nu \right] = \mathcal{O}(\rho^\ell). \quad (29)$$

(II) From part (I), it follows that $(\sigma_i^2(t); i \in \mathbb{Z})$ and $(y_i(t); i \in \mathbb{Z})$ are weakly dependent processes.

Corollary 2.2 part (I) expresses geometric L^ν -m-approximability with respect to the L^2 -norm. The concept of L^ν -m-approximability for functional data was first introduced by Hörmann and Kokoszka (2010). A process that is geometrically L^ν -m-approximable can be approximated by an m-dependent process at a geometric rate. A stochastic process (X_n) is called m-dependent if observations that are more than m time steps apart are independent. More precisely, for any time index k , the collection of random variables $\{X_n : n \leq k\}$ is independent of the collection $\{X_n : n \geq k + m\}$. In other words, the past up to time k has no influence on the future starting from time $k + m$. Formally, X_n is m-dependent if, for any integer k , the σ -algebra $\mathcal{F}_k^- = \sigma\{\dots, X_{k-2}, X_{k-1}, X_k\}$ generated by the past observations is independent of the σ -algebra generated by the future observations $\mathcal{F}_{k+m}^+ = \sigma\{X_{k+m}, X_{k+m+1}, X_{k+m+2}, \dots\}$. Aue et al. (2017) uses L^ν -m-approximability to show that the empirical eigenfunctions computed from the empirical covariance kernel of $y_i^2(t)$ converge to the corresponding eigenfunctions of the true covariance kernel of $y_i^2(t)$. The use of this result will be discussed in Section 2.3. They also establish that, if equation (29) holds for some $\nu \geq 2$, this weak-dependence condition is strong enough to derive a functional central limit theorem; see Berkes et al. (2013) and Jirak (2013) for background on the weak invariance principles underlying such results.

Finally, Kühnert (2020) extends the L^ν -m-approximability result in equation (29) to FGARCH processes of arbitrary orders p and q , as well as to L^p spaces with $p \in [1, \infty)$, and more generally to separable Banach spaces of functions on $[0, 1]$ endowed with the supremum norm and closed under pointwise multiplication.

2.3 Estimation

In this section, the estimation of an FGARCH(1, 1) model will be discussed. I consider a setting in which a finite sample of N functional observations $\{y_i(t)\}_{i \in \{1, \dots, N\}}$ is available. Although functional data are commonly recorded on a dense discrete grid in practical applications, I assume here that the underlying curves are directly observed or can be reliably reconstructed from the data.

Because the objects to be estimated are infinite-dimensional, direct estimation from a finite sample is generally ill-posed and therefore requires some form of dimension reduction. For this reason, Aue et al. (2017) imposes a standard finite-dimensional approximation on the intercept function $\delta(t)$ and on the integral operators α and β . In particular, they assume that all three objects can be approximated using a fixed number M of orthonormal basis functions $\Phi_M = \{\varphi_1(t), \dots, \varphi_M(t)\}$ defined on $[0, 1]$. Under this assumption, $\delta(t)$ is represented as a finite linear combination of these basis functions, while the operators α and β are represented through their kernel functions using the same basis:

$$\delta(t) = \sum_{m=1}^M d_m \varphi_m(t), \quad (30)$$

$$\alpha(t, s) = \sum_{m, m'=1}^M a_{m, m'} \varphi_m(t) \varphi_{m'}(s), \quad (31)$$

$$\beta(t, s) = \sum_{m, m'=1}^M b_{m, m'} \varphi_m(t) \varphi_{m'}(s). \quad (32)$$

From this representation, it is easy to see that estimating the functional objects $\delta(t)$, α and β simplifies to estimating $\{d_1, \dots, d_M, a_{1,1}, \dots, a_{M,M}, b_{1,1}, \dots, b_{M,M}\}$. To further simplify notation, Aue et al. (2017) defines the following quantities:

$$\mathbf{d} = \begin{pmatrix} d_1 \\ \vdots \\ d_M \end{pmatrix}; \quad \mathbf{A} = \begin{pmatrix} a_{1,1} & \dots & a_{1,M} \\ \vdots & \ddots & \vdots \\ a_{M,1} & \dots & a_{M,M} \end{pmatrix}; \quad \mathbf{B} = \begin{pmatrix} b_{1,1} & \dots & b_{1,M} \\ \vdots & \ddots & \vdots \\ b_{M,1} & \dots & b_{M,M} \end{pmatrix} \quad (33)$$

To estimate $\mathbf{d}$, $\mathbf{A}$, and $\mathbf{B}$, the squared observations $(y_i^2(t))_{i \in \{1, N\}}$ and conditional volatility functions $(\sigma_i^2(t))_{i \in \{1, N\}}$ are projected onto Φ_M . This yields the M -dimensional vectors $\mathbf{y}_i^{(2)} = (y_{i,1}^{(2)} \dots y_{i,M}^{(2)})^\top$ and $\mathbf{s}_i^{(2)} = (s_{i,1}^{(2)} \dots s_{i,M}^{(2)})^\top$ whose components are given by the inner products with the basis functions in Φ_M $y_{i,m}^{(2)} = \langle y_i^2, \varphi_m \rangle$, $s_{i,m}^{(2)} = \langle \sigma_i^2, \varphi_m \rangle$. Under

the finite-dimensional parametrisation introduced above, the volatility recursion can then be written in terms of the projected coefficient vectors as

$$\mathbf{s}_i^{(2)} = \mathbf{d} + \mathbf{A}\mathbf{y}_{i-1}^{(2)} + \mathbf{B}\mathbf{s}_{i-1}^{(2)}. \quad (34)$$

Solving this equation recursively obtains an infinite-order representation for $\mathbf{s}_i^{(2)}$ that only depends on $\mathbf{y}_i^{(2)}$

$$\mathbf{s}_i^{(2)} = \sum_{\ell=1}^{\infty} \mathbf{B}^{\ell-1} \mathbf{d} + \sum_{\ell=1}^{\infty} \mathbf{B}^{\ell-1} \mathbf{A} \mathbf{y}_{i-\ell}^{(2)}. \quad (35)$$

Since the true parameter vector $\theta = (\mathbf{d}^\top, \mathbf{A}^\top, \mathbf{B}^\top)^\top$ is unknown, equation (35) is expressed at an arbitrary parameter value $\underline{\theta} = (\underline{\mathbf{d}}^\top, \underline{\mathbf{A}}^\top, \underline{\mathbf{B}}^\top)^\top \in \Theta \subset \mathbb{R}^{M+2M^2}$

$$\tilde{\mathbf{s}}_i^{(2)} = \tilde{\mathbf{s}}_i^{(2)}(\underline{\theta}) = \sum_{\ell=1}^{\infty} \underline{\mathbf{B}}^{\ell-1} \underline{\mathbf{d}} + \sum_{\ell=1}^{\infty} \underline{\mathbf{B}}^{\ell-1} \underline{\mathbf{A}} \mathbf{y}_{i-\ell}^{(2)}. \quad (36)$$

Since in practice only a finite time frame of the series can be observed, $\mathbf{s}_i^{(2)}$ is approximated by $\hat{\mathbf{s}}_i^{(2)}$

$$\hat{\mathbf{s}}_i^{(2)} = \hat{\mathbf{s}}_i^{(2)}(\underline{\theta}) = \sum_{\ell=1}^{i-1} \underline{\mathbf{B}}^{\ell-1} \underline{\mathbf{d}} + \sum_{\ell=1}^{i-1} \underline{\mathbf{B}}^{\ell-1} \underline{\mathbf{A}} \mathbf{y}_{i-\ell}^{(2)}. \quad (37)$$

The objective, thus, has to be to choose an estimator $\hat{\underline{\theta}}_n$ such that $\hat{\mathbf{s}}_i^{(2)}$ is close to $\mathbf{s}_i^{(2)}$. Since $\mathbf{s}_i^{(2)}$ is unobservable, Aue et al. (2017) imposes

$$\mathbb{E}[\varepsilon_i^2(t)] = 1; \quad \forall t \in [0, 1]. \quad (38)$$

This is without loss of generality since this condition can always be fulfilled by an appropriate way of scaling. Under this condition, $\mathbb{E}[y_i^2(t)|\mathcal{F}_{i-1}] = \sigma_i^2(t)$ and therefore, after projection onto Φ_M , $\mathbb{E}[y_i^{(2)}|\mathcal{F}_{i-1}] = s_i^{(2)}$ where $\mathcal{F}_{i-1}$ is the sigma-algebra generated by all $\varepsilon_\ell : \ell < i$, $\mathcal{F}_{i-1} = \sigma(\varepsilon_\ell : \ell < i)$. As a consequence, Aue et al. (2017) defines the following LS estimator

$$\hat{\underline{\theta}}_n = \arg \min \{S_n(\underline{\theta}) = \sum_{i=2}^n \{y_i^{(2)} - \hat{s}_i^{(2)}(\underline{\theta})\}^\top \{y_i^{(2)} - \hat{s}_i^{(2)}(\underline{\theta})\} : \underline{\theta} \in \Theta\}. \quad (39)$$

In the following, I discuss the strong consistency and asymptotic distribution of the LS

estimator $\hat{\underline{\theta}}_n$. The required assumptions are summarized below in a slightly condensed form following Aue et al. (2017).

Assumption 2.1. It is assumed that

(A1) $\mathbf{y}_1^{(2)}$ is not measurable with respect to $\mathcal{F}_0$,

(A2) Θ is a compact set and θ is in the interior of Θ and

(A3) there are $0 < c_1$ and $c_2 < 1$ such that $c_1 \leq |\det(\underline{\mathbf{A}})|$ and $\|\underline{\mathbf{B}}\| < 1$ for all $\underline{\theta} = (\underline{\mathbf{d}}^\top, \underline{\mathbf{A}}^\top, \underline{\mathbf{B}}^\top) \in \Theta$.

Assumption (A2) is a standard compactness and interiority condition on the parameter space. Assumption (A3) imposes uniform identification and stability restrictions over Θ . Consequently, it guarantees that $\underline{\mathbf{A}}$ is non-singular and that $\underline{\mathbf{B}}$ is uniformly contractive for all $\underline{\theta} \in \Theta$. In particular, these properties hold at the true parameter value θ .

Using this assumption, the strong consistency of $\hat{\underline{\theta}}_n$ is established in the following Theorem.

Theorem 2.3. If $\{y_i(t)\}_{i \in \mathbb{Z}}$ follows a FGARCH(1, 1) process as defined in equations (18) and (19), $\delta(t)$ is a non-negative function, and α and β are integral operators that map non-negative functions to non-negative functions, equation (24) holds with $\nu = 1$, $\mathbb{E} [\|\varepsilon_0\|_2^2] < \infty$, and equation (38) holds, then under Assumption 2.1 $\hat{\underline{\theta}}_n$ is strongly consistent, meaning that

$$\hat{\underline{\theta}}_n \xrightarrow{\text{a.s.}} \theta \quad (40)$$

as $n \rightarrow \infty$.

The full proof of Theorem 2.3 can be found in Aue et al. (2017).

To examine the asymptotic distribution of $\hat{\underline{\theta}}_n$, the following quantities are useful

$$\mathbf{H}_0 = \mathbb{E} \left[\frac{\partial \tilde{\mathbf{s}}_0^{(2)}(\theta)}{\partial \underline{\theta}} \right], \quad \mathbf{J}_0 = \mathbb{E} \left[\left\{ \mathbf{y}^{(2)} - \mathbf{s}_0^{(2)} \right\} \left\{ \mathbf{y}^{(2)} - \mathbf{s}_0^{(2)} \right\}^\top \right], \quad (41)$$

$$\mathbf{Q}_0 = \mathbb{E} \left[\left(\frac{\partial \tilde{\mathbf{s}}_0^{(2)}(\theta)}{\partial \underline{\theta}} \right)^\top \left(\frac{\partial \tilde{\mathbf{s}}_0^{(2)}(\theta)}{\partial \underline{\theta}} \right) \right]. \quad (42)$$

Under the conditions stated in Theorem 2.3, the matrices $\mathbf{H}_0$ and $\mathbf{Q}_0$ are well defined. Moreover, if $\mathbb{E}[\|\sigma_0^2\|_2^4] < \infty$ and $\mathbb{E}[\|\varepsilon_0^2\|_2^4] < \infty$, the matrix $\mathbf{J}_0$ exists. Here, $\mathbf{H}_0$ is the expected first-derivative matrix of the fitted volatility coefficients with respect to the parameter vector, while $\mathbf{J}_0$ is the covariance matrix of the residual term $\mathbf{y}_0^{(2)} - \mathbf{s}_0^{(2)}$. Thus, $\mathbf{H}_0$ captures the local sensitivity of the model with respect to the parameters, whereas $\mathbf{J}_0$ measures the variability of the part of the projected squared observations that is not explained by the conditional volatility. The matrix $\mathbf{Q}_0$ is the second-moment matrix of the derivative of $\tilde{\mathbf{s}}_0^{(2)}(\theta)$ with respect to the parameter vector. In this sense, it plays the role of a curvature matrix for the LS criterion. More specifically, as shown in Section A.4, the second derivative of the LS objective converges to $2\mathbf{Q}_0$ in the proof of the asymptotic result. Following the standard theory of M -estimators, it is therefore additionally assumed that $\mathbf{Q}_0$ is non-singular. This guarantees invertibility and ensures that the objective function is locally well behaved around the true parameter.

Theorem 2.4. Assume that the conditions of Theorem 2.3 are satisfied, that $\mathbf{Q}_0$ is non-singular, and that $\mathbb{E}[\|y_0^2\|_2^4] < \infty$. Then, as $n \rightarrow \infty$,

$$\sqrt{n}(\hat{\underline{\theta}}_n - \theta) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \mathbf{Q}_0^{-1} \mathbf{H}_0^\top \mathbf{J}_0 \mathbf{H}_0 \mathbf{Q}_0^{-1}), \quad (43)$$

where the limiting distribution is a $(2M^2 + M)$ -dimensional normal distribution.

The proof of Theorem 2.4 relies on a componentwise application of the mean value theorem. Define

$$\hat{Z}_n(\underline{\theta}) = \frac{1}{n} \sum_{i=2}^n \left\{ y_i^{(2)} - \hat{s}_i^{(2)}(\underline{\theta}) \right\}^\top \left\{ y_i^{(2)} - \hat{s}_i^{(2)}(\underline{\theta}) \right\},$$

the sample LS criterion for a generic parameter vector $\underline{\theta}$. Denote its gradient and Hessian by $\frac{\partial \hat{Z}_n(\underline{\theta})}{\partial \underline{\theta}}$ and $\frac{\partial^2 \hat{Z}_n(\underline{\theta})}{\partial \underline{\theta}^2}$, respectively. Applying the mean value theorem componentwise to the gradient of $\hat{Z}_n$ between the true parameter θ and the estimator $\hat{\underline{\theta}}_n$ yields

$$\frac{\partial \hat{Z}_n(\hat{\underline{\theta}}_n)}{\partial \underline{\theta}} - \frac{\partial \hat{Z}_n(\theta)}{\partial \underline{\theta}} = \frac{\partial^2 \hat{Z}_n(\check{\theta})}{\partial \underline{\theta}^2} (\hat{\underline{\theta}}_n - \theta),$$

for some intermediate value $\check{\theta}$ lying between θ and $\hat{\underline{\theta}}_n$. Since $\hat{\underline{\theta}}_n$ minimizes the criterion, its gradient vanishes, so that

$$\sqrt{n}(\hat{\underline{\theta}}_n - \theta) = - \left(\frac{\partial^2 \hat{Z}_n(\check{\theta})}{\partial \underline{\theta}^2} \right)^{-1} \sqrt{n} \frac{\partial \hat{Z}_n(\theta)}{\partial \underline{\theta}}.$$

The asymptotic distribution of $\hat{\underline{\theta}}_n$ therefore follows from the limiting behavior of the gradient and the Hessian. More precisely, a uniform law of large numbers implies that the Hessian converges to $2\mathbf{Q}_0$, while a central limit theorem for orthogonal martingale differences yields the asymptotic distribution of the gradient evaluated at θ . Combining these two results gives the asymptotic normality stated in Theorem 2.4. The full proof is deferred to Section A.4.

Cerovecki et al. (2019) proposes an alternative estimator based on QML estimation. Applying maximum likelihood techniques to functional data is generally not straightforward since a genuine likelihood for curves is typically not available since there are no densities for infinite-dimensional data. As a consequence, Cerovecki et al. (2019) also assumes that $\delta(t)$, α , and β can be decomposed into a finite combination of baseline functions similar to equations (30-32). Under this parametrisation, the projected squared process satisfies a recursion analogous to equation (34), which can be related to a semi-strong multivariate CCC-GARCH model with $R = \mathbb{I}_M$, where $\mathbb{I}_M$ denotes the $M \times M$ identity matrix.

Cerovecki et al. (2019) first establishes consistency and asymptotic normality for the QML estimator in this semi-strong multivariate CCC-GARCH setting and then use these results to derive consistency and asymptotic normality for the corresponding pseudo-QML estimator of the FGARCH model. The estimator is defined by

$$\hat{\underline{\theta}}_n = \arg \min_{\theta \in \Theta} \sum_{i=1}^n \tilde{\ell}_i(\theta), \quad \tilde{\ell}_i(\theta) = \sum_{m=1}^M \left\{ \frac{\langle y_i^2, \phi_m \rangle}{\langle \tilde{\sigma}_i^2, \phi_m \rangle} + \log \langle \tilde{\sigma}_i^2, \phi_m \rangle \right\}, \quad (44)$$

where the fitted volatility is calculated recursively as

$$\tilde{\sigma}_i^2 = \tilde{\sigma}_i^2(\theta) = \delta + \alpha y_{i-1}^2 + \beta \tilde{\sigma}_{i-1}^2, \quad \text{for } i = 1, \dots, n. \quad (45)$$

Finally, Aue et al. (2017) discusses the choice of basis functions. In principle, any orthonormal system on $[0, 1]$ can be used as basis functions. Common examples include B-splines, Fourier bases, and wavelet functions. As a data-driven alternative, one may estimate functional principal components (FPCs) from the squared observations and use these as basis functions. To do so, define the covariance kernel of $y_0^2(t)$ as

$$D(t, s) = \text{Cov}(y_0^2(t), y_0^2(s)), \quad s, t \in [0, 1]. \quad (46)$$

An estimate of $D(t, s)$ can then be obtained by

$$\hat{D}(t, s) = \frac{1}{n} \sum_{i=1}^n \{y_i^2(t) - \bar{y}_n^{(2)}(t)\} \{y_i^2(s) - \bar{y}_n^{(2)}(s)\} \quad (47)$$

where $\bar{y}_n^{(2)}(t) = \frac{1}{n} \sum_{i=1}^n y_i^2(t)$. Using the spectral decomposition of $\hat{D}(t, s)$ the empirical eigenfunctions $\{\hat{\varphi}_1, \dots, \hat{\varphi}_M\}$ and eigenvalues $\{\hat{\lambda}_1, \dots, \hat{\lambda}_M\}$ are obtained. Note that under equation (38) $\mathbb{E}[y_i^2(t) | \mathcal{F}_{i-1}] = \sigma_i^2(t)$. Hence, the covariance structure of the squared observations contains information about the latent volatility process. This suggests using the leading empirical eigenfunctions of $\hat{D}$ as a data-driven basis, since they capture the dominant directions of variation in y_i^2 and therefore also in the unobservable volatilities. Aue et al. (2017) shows that, if the empirical eigenfunctions are used in place of the deterministic basis Φ_M and $\int \mathbb{E}[y_0^4(t)] dt < \infty$ then the strong consistency result in Theorem 2.3 remains valid. This extension is justified by the weak-dependence properties of the FGARCH(1, 1) process, in particular its geometric L^2 - m -approximability established earlier.

The same basis functions cannot be used in this exact way when using the QML estimator since the QML requires the basis functions to be non-negative. This ensures that the inner products in equation (44) are positive and thus the minimization problem is well defined and reaches its minimum on the compact set Θ .

Cerovecki et al. (2019) therefore proposes to use families of non-negative and linearly independent functions, such as the power basis, the exponential basis, or polynomial bases that are non-negative on $[0, 1]$. Alternatively, they develop a data-driven method based on Functional Principal Component Analysis (FPCA). Since standard PCs are orthonormal and therefore need not be non-negative, they cannot be used directly in the pseudo-QML framework. Cerovecki et al. (2019) proposes to shift the eigenfunctions obtained from FPCA using the following transformation. First, define $\phi_1(t) = 1$ and $\phi_2(t) = \mu(t)$. Then, define the remaining basis functions by

$$\phi_m(t) := \varphi_{m-2}(t) - \inf_{t \in [0, 1]} \varphi_{m-2}(t) \wedge 0, \quad \text{for all } m = 3, \dots, M \quad (48)$$

where $\wedge$ denotes the minimum operator and $\inf_{t \in [0, 1]} \varphi_{m-2}(t)$ is the minimum value of φ_{m-2} over $[0, 1]$. This transformation shifts each PC upward, if necessary, so that the resulting function is non-negative on $[0, 1]$. In practice, the mean function and eigenfunctions are

replaced by their empirical counterparts. Cerovecki et al. report in their simulation study that this empirical, data-driven choice performs relatively well, even when compared with settings in which the true, but unobserved, basis functions from the data-generating process are used.

Using a set of linearly independent functions instead of orthonormal eigenfunctions comes at the cost of losing some of the structural advantages of an orthonormal basis. In particular, because of the loss of orthogonality, the basis representation does not retain the same computational simplicity and interpretability as in the case of FPCs. Moreover, the associated Gram matrix is, in general, no longer equal to the identity, which may complicate the finite-dimensional representation of the model and may lead to numerical instability if the basis functions are highly correlated. These issues have to be balanced against the non-negativity requirement of the QML approach, which makes this choice of basis functions necessary. The practical relevance of this trade-off will be discussed further in Section 3, where the comparison of the QML and LS estimators, in particular the results in Section 3.2, may also reflect the effect of this choice of basis functions.

2.4 Value-at-risk

This subsection provides a brief overview of the concept of VaR. After a short historical overview is given, the subsection turns to the formal definition of VaR, its main theoretical properties, and the central criticisms that have been raised against it.

VaR emerged as an operational tool for market risk measurement in the early 1990s, when J.P. Morgan introduced it internally as a standardized approach to measuring firm-wide market risk. The public release of the *RiskMetrics* methodology in 1994 contributed to the widespread adoption of VaR by financial institutions. VaR was subsequently incorporated into financial regulation through the 1996 Market Risk Amendment to the Basel I Accord, which allowed banks to use internal VaR models for the calculation of regulatory capital. Early contributions to the measurement of downside risk can be found in Leavens (1945), while related concepts appear in the safety-first criterion of Roy (1952) and the mean-variance framework of Markowitz (1952). However, it should be noted that these works did not define VaR in its modern form.

Intuitively, VaR is intended to specify a threshold below which returns fall only with a predetermined probability. In probability terms, this means that, under the assumption that the map $x \mapsto \mathbb{P}(y_i \leq x \mid \mathcal{F}_{i-1})$ is strictly increasing, where $\mathcal{F}_{i-1}$ denotes the σ -algebra containing the information available up to time $i - 1$, it holds that

$$\mathbb{P}(y_i \leq \text{VaR}_i^\tau \mid \mathcal{F}_{i-1}) = \tau, \quad \tau \in (0, 1). \quad (49)$$

Here, VaR_i^τ denotes the almost surely unique $\mathcal{F}_{i-1}$ -measurable random variable corresponding to the quantile level τ . In the intraday functional setting considered in this thesis, this definition is applied pointwise in t . More precisely, the intraday VaR curve is given by $\text{VaR}_i^\tau(t)$, where for each $t \in [0, 1]$ it holds that

$$\mathbb{P}(y_i(t) \leq \text{VaR}_i^\tau(t) \mid \mathcal{F}_{i-1}) = \tau. \quad (50)$$

Two important properties of VaR follow from this definition. First, VaR is not subadditive, meaning that the VaR of a combined portfolio can exceed the sum of the individual VaRs of the portfolios from which it is formed. As a consequence, VaR is not a coherent risk measure (Artzner et al., 1999). To see this in a simple scalar example, let A and B be two independent random variables with identical distributions given by

$$\mathbb{P}(A = 0) = 0.95, \quad \mathbb{P}(A = -10) = 0.05, \quad (51)$$

and

$$\mathbb{P}(B = 0) = 0.95, \quad \mathbb{P}(B = -10) = 0.05. \quad (52)$$

Then, at the confidence level $\tau = 0.05$, the VaR of each individual random variable is equal to 0. However, for the sum $A + B$, the probability of obtaining the value 0 is

$$\mathbb{P}(A = 0 \text{ and } B = 0) = 0.95 \cdot 0.95 = 0.9025. \quad (53)$$

Hence, the lower 5% quantile of $A + B$ is -10 , so that the VaR of the combined position is -10 .

A second important property of VaR is that it only captures the quantile at the chosen confidence level and therefore does not reflect the magnitude of losses beyond this

threshold.

These two features, together with the widespread use of VaR as a risk management tool, have led to substantial criticism of VaR. As a result, extensions to VaR have been established. The most important extension to VaR is Expected Shortfall, also called Conditional VaR. Expected Shortfall in the intraday setting considered here is defined as follows:

$$\text{ES}_i^\tau(t) = \mathbb{E}[y_i(t) | \mathcal{F}_{i-1}, y_i(t) < \text{VaR}_i^\tau(t)]. \quad (54)$$

Expected Shortfall is subadditive and takes the behavior of the tail of the distribution into account. Therefore, Expected Shortfall addresses the two main theoretical issues of VaR. Rice et al. (2020a) also considers Expected Shortfall in the appendix of their paper. They show that the method explained in Section 3 can be extended to the concept of Expected Shortfall. This is beyond the scope of this thesis.

3 Simulation Study

In the next section, I will explore the finite sample properties of the FGARCH model and highlight a few practical considerations that must be made when using the FGARCH model for forecasting intra-day VaR in practice.

First, it should be emphasized that functional data are, in practice, observed on a finite grid. Consequently, both the properties discussed in Section 2.2 and the estimation procedure described in Section 2.3 are implemented in a discretized setting. This discretization does not yield a purely multivariate GARCH model; rather, it leads to a finite-dimensional approximation of the underlying functional model. In particular, operations defined in the functional setting, such as the integrals appearing in inner products, must be approximated numerically, which requires explicit consideration of the grid structure and the distances between adjacent grid points. For all simulations, functional data are approximated on an equidistant grid between $[0, 1]$ with 100 grid points.

Second, forecasting VaR requires an approximation of the pointwise distribution of the innovation process $\varepsilon_i(t)$. The resulting pointwise VaR forecast is given by $\hat{\sigma}_i(t)\hat{\varepsilon}_i^\tau(t)$, where

$\hat{\sigma}_i(t)$ denotes the forecast of $\sigma_i(t)$ obtained from the fitted FGARCH model and $\hat{\varepsilon}_i^\tau(t)$ denotes the estimated pointwise τ -quantile of $\varepsilon_i(t)$. To estimate $\hat{\varepsilon}_i^\tau(t)$, Rice et al. (2020a) proposes three methods. The first assumes that $\varepsilon_i(t)$ follows an Ornstein-Uhlenbeck process. The second estimates $\hat{\varepsilon}_i^\tau(t)$ by means of a bootstrap algorithm. The third uses extreme value theory to obtain a better approximation of the tail behaviour. In what follows, I focus on the second method, as it performed best in the empirical study of Rice et al..

To obtain $\hat{\varepsilon}_i^\tau(t)$ using the bootstrap algorithm, first the fitted $\hat{\sigma}_i(t)$ needs to be computed. Next, $\hat{\varepsilon}_i(t)$ is calculated by $\hat{\varepsilon}_i(t) = \frac{y_i(t)}{\hat{\sigma}_i(t)}$. Finally, to get $\hat{\varepsilon}_i^\tau(t)$ the estimated error terms are resampled B many times with replacement. The estimator $\hat{\varepsilon}_i^\tau(t)$ is then defined as the pointwise τ -quantile of the bootstrap sample $(\hat{\varepsilon}_i^{(b)}(t))_{b \in [1, B]}$.

Third, to evaluate the quality of the VaR forecasts, I employ the following backtesting procedure. To assess the accuracy of the intraday VaR forecasts, I test whether the conditional coverage equals the nominal level τ . The null hypothesis is thus

$$\mathcal{H}_0 : \mathbb{E}[Z_i^\tau(t) - \tau] = 0 \text{ for all } t \in [0, 1] \quad (55)$$

. To test $\mathcal{H}_0$, define the pointwise violation process $Z_i^\tau(t)$ at quantile τ

$$Z_i^\tau(t) = \mathbb{1}_{\{y_i(t) \leq \widehat{\text{VaR}}_i^\tau(t)\}}; \quad i \in \{1, \dots, N\} \quad t \in [0, 1] \quad (56)$$

where $\mathbb{1}$ denotes the indicator function and $\widehat{\text{VaR}}_i^\tau(t)$ is the pointwise VaR forecast. Rice et al. (2020a) then proposes the following test statistic:

$$T_n = N \|\bar{Z} - \tau\|^2. \quad (57)$$

Here, $\bar{Z}$ denotes the sample mean function of the violation process $\bar{Z} = \frac{1}{N} \sum_{i=1}^N Z_i^\tau(t)$. Under the additional simplifying assumption that the violation process is i.i.d. across i , Rice et al. (2020a) shows that

$$T_n \xrightarrow{\mathcal{D}} \sum_{\ell=1}^{\infty} \lambda_\ell \mathcal{N}_\ell^2 \stackrel{\mathcal{D}}{=} \Xi_1, \quad N \rightarrow \infty, \quad (58)$$

where $(\mathcal{N}_\ell : \ell \geq 1)$ are i.i.d. standard normal distributed random variables, and $(\lambda_\ell : \ell \geq 1)$ are the eigenvalues of the covariance operator of the violation process, or more specif-

ically, the eigenvalues of the kernel integral operator. The kernel of the integral operator is defined by $C_Z(t, s) = \text{Cov}(Z_0(t), Z_0(s))$, $t, s \in [0, 1]$. One can then approximate the distribution of Ξ_1 via the use of Monte Carlo simulation after estimating $(\lambda_\ell : \ell \geq 1)$ from the sample version of the covariance operator. The corresponding p-value is then approximated by $p = \mathbb{P}(T_n < \Xi_1)$. Additional tests for evaluating the performance of the estimators and parameters chosen are introduced where needed.

Fourth, another aspect concerns the choice of the initial value for both the data-generating process and the estimation procedure. For the simulations, I initialize the conditional variance curve by setting $\sigma_1^2(t) = \delta(t)$. For the estimation step, different choices have been used in the literature. In the LS framework, Aue et al. (2017) considers an initialization based on the projected volatility recursion, while Cerovecki et al. (2019) uses the average of the squared curves from the first trading week, that is, $\frac{1}{5} \sum_{i=1}^5 y_i^2(t)$, as the initial value. In this thesis, I adopt a different approach. In analogy with the standard univariate GARCH model, I initialize the recursion at its unconditional mean whenever this quantity exists. More precisely, for the LS estimator I set $\mathbf{s}_2^{(2)} = (\mathbf{I} - \mathbf{A} - \mathbf{B})^{-1}\mathbf{d}$, and for the QML estimator I use $\sigma_1^2(t) = (\mathcal{I} - \alpha - \beta)^{-1}\delta(t)$. Here, $\mathbf{I}$ and $\mathcal{I}$ denote the identity matrix and the identity operator, respectively. Under the conditions ensuring existence of the unconditional mean, these initializations are the functional analogues of the unconditional variance initialization commonly used in the univariate GARCH literature.

For all simulations, I consider the VaR levels $\tau \in \{0.025, 0.01, 0.005\}$. In addition, each simulated sample is divided into a burn-in, a training, and an evaluation period. The burn-in length is set to $N = 1000$, and these observations are discarded. This reduces the effect of the initial value and brings the process close to its stationary regime before the training sample begins. The evaluation sample size is fixed at $N = 250$ throughout all simulations. All reported simulation results are computed on the evaluation sample. I use one-day-ahead forecasts.

3.1 Simulation 1

For the first simulation, I adopt a design closely related to the first simulation setup in Aue et al. (2017) and the first example in Cerovecki et al. (2019). Set $\alpha(t, s) = \beta(t, s) = 12t(1 - t)s(1 - s)$. In both references, the innovation functions are specified through

Gaussian-process constructions of Ornstein-Uhlenbeck type. Cerovecki et al. (2019) then shows that under this setting, their QML estimator outperforms the LS estimator by Aue et al.. Such Gaussian innovation specifications may be restrictive for financial data, since intraday returns are often found to exhibit heavy tails. The empirical high-frequency literature does not support the idea that intraday returns are well approximated solely by Brownian-type innovations. Instead, many studies document excess kurtosis, fat tails, or power-law tails in intraday equity and foreign exchange returns, and some even find evidence against the existence of a finite fourth moment at intraday frequencies (Loretan and Phillips, 1994; Gopikrishnan et al., 1999; Plerou et al., 1999; Degiannakis et al., 2023). This provides a natural motivation for complementing Brownian-motion-based simulation designs with heavy-tailed innovations. To test the results obtained in Cerovecki et al. (2019) using an alternative specification for the random innovation functions, I use the following specification for all $\varepsilon_i(t)$. Let $0 \leq t_1 < \dots < t_m \leq 1$ denote the simulation grid, and define

$$\Delta_1 = e^{t_1}, \quad \Delta_j = e^{t_j} - e^{t_{j-1}}, \quad j = 2, \dots, m. \quad (59)$$

Further, let $U_{i1}, \dots, U_{im}$ be independent Student- t random variables with ν degrees of freedom, and standardize them by

$$\xi_{ij} = \sqrt{\frac{\nu-2}{\nu}} U_{ij}, \quad j = 1, \dots, m, \quad (60)$$

so that $\mathbb{E}[\xi_{ij}] = 0$ and $\text{Var}(\xi_{ij}) = 1$ for $\nu > 2$. I then define the partial-sum process

$$B_i^\nu(t_j) = \sum_{k=1}^j \sqrt{\Delta_k} \xi_{ik}, \quad j = 1, \dots, m, \quad (61)$$

and construct the innovation function on the grid by

$$\varepsilon_i(t_j) = e^{-t_j/2} B_i^\nu(t_j), \quad j = 1, \dots, m. \quad (62)$$

This mirrors the Ornstein-Uhlenbeck-type construction used in the Gaussian case, but replaces Gaussian increments by standardized Student- t increments. By construction, $\text{Var}(\varepsilon_i(t_j)) = 1$ for every grid point t_j , so that equation (38) holds on the simulation grid. Since a Student- t variable has a finite second (fourth) moment if and only if $\nu > 2$ ($\nu > 4$), the same condition is sufficient and necessary for the innovation values to

possess finite second (fourth) moments at each grid point. Additionally, I chose a different specification for δ . To ensure that δ , α , and β lie in the chosen one-dimensional basis representation, I set δ equal to $\delta(t) = 0.01\sqrt{30}t(1-t)$. In addition, I assume that the true basis function $\sqrt{30}t(1-t)$ is known. Under this specification, the true parameter vector is $\theta = \begin{pmatrix} d = 0.01 & a = 0.4 & b = 0.4 \end{pmatrix}$. The simulations were run for three values of ν , $\nu = \{3, 5, 25\}$. Consequently, for $\nu = 3$ the innovation values have finite second moments but no finite fourth moments. Hence, the fourth-moment condition in Theorem 2.4 is violated in this case. In addition, three training sample sizes were considered for the simulations to test the effect of larger samples.

Table 1: Mean bias and standard deviation of estimators for LS and QML estimation

Sample size	$\nu = 3$			$\nu = 5$			$\nu = 25$		
	d	a	b	d	a	b	d	a	b
LS Estimation									
250	.0211 (.0991)	-.1194 (.1526)	-.2293 (.2621)	.0087 (.0101)	-.1039 (.1051)	-.1132 (.1772)	.0072 (.0114)	-.0744 (.1017)	-.0617 (.1525)
500	.0109 (.0158)	-.1223 (.1446)	-.1251 (.2832)	.0070 (.0084)	-.0777 (.1148)	-.0584 (.1898)	.0048 (.0040)	-.0679 (.0792)	-.0446 (.1078)
1000	.0149 (.0606)	-.1175 (.1250)	-.1363 (.2288)	.0056 (.0064)	-.0547 (.1498)	-.0672 (.1946)	.0035 (.0035)	-.0659 (.0695)	-.0089 (.1031)
QML Estimation									
250	.0026 (.0080)	-.0371 (.1866)	-.0668 (.1966)	.0008 (.0057)	-.0106 (.1477)	-.0244 (.1703)	.0009 (.0040)	.0136 (.1048)	-.0285 (.1182)
500	.0006 (.0048)	.0205 (.1919)	-.0303 (.1799)	.0010 (.0038)	.0166 (.1118)	-.0237 (.1413)	.0009 (.0037)	-.0068 (.0793)	-.0189 (.1153)
1000	.0011 (.0045)	-.0040 (.1151)	-.0560 (.1259)	.0006 (.0024)	-.0065 (.0772)	-.0089 (.0859)	.0005 (.0023)	-.0007 (.0611)	-.0068 (.0819)

Table 1 reports the mean bias and standard deviation of the LS and QML estimators across all combinations of sample size and ν . Two main findings emerge from these results: First, heavier tails have a pronounced negative effect on estimation accuracy, and this effect is particularly strong for LS estimation. The contrast between $\nu = 3$ and $\nu \in \{5, 25\}$ is especially stark for LS, where both the mean bias and the standard deviation increase substantially. This deterioration is particularly visible for the estimates of d and b . Second, QML generally outperforms LS estimation in terms of both mean bias and standard deviation, which is consistent with the findings of Cerovecki et al. (2019). This advantage is particularly pronounced in the case $\nu = 3$, where LS estimation deteriorates sharply while QML remains comparatively stable. These results suggest that the absence

of finite fourth moments mainly affects the finite-sample stability of the estimators, with LS estimation appearing substantially more sensitive than QML estimation. In addition, despite some irregularities, larger sample sizes generally lead to more accurate estimation, as reflected in lower bias and dispersion for most parameter combinations.

A similar overall pattern is observed for the backtesting results based on the p-values, although the difference between LS and QML is less pronounced there than in the bias and dispersion measures. Here as well, QML appears to perform better than LS on average, and forecast performance tends to improve with larger sample sizes and thinner-tailed innovation specifications. I defer the table containing p-values to Section A.5.

3.2 Simulation 2

In the second simulation, I consider a slightly more complex setup. First, I depart from the finite-dimensional basis specification imposed in equations (30–32) by choosing alternative functional forms for δ , α and β . To do so, I keep $\beta(t, s) = 12t(1 - t)s(1 - s)$, but replace α by a Gaussian kernel defined as $\alpha(t, s) = 0.5 \exp(-(t - s)^2)$ and set $\delta(t)$ to $\exp(-t^2)$.

Furthermore, to compare LS and QML estimation in a setting with more than one retained basis direction, I consider two values of M for each method. For both LS and QML estimation, I use the respective data-driven basis construction described in Section 2.3. This also allows me to examine the effect of using orthonormal basis functions in the LS approach versus non-negative basis functions that are only required to be linearly independent in the QML approach. For LS estimation I consider the following values for M , $M \in \{2, 4\}$. I find that two FPCs explain about 90% of the variance in y_i^2 while four FPCs explain about 95%. In the QML approach, the data-driven basis construction fixes the first two baseline functions as a constant function and the sample mean of $y_i^2(t)$. I therefore consider $M \in \{4, 6\}$. This makes the effective number of additional data-driven directions more comparable across the two estimation methods. Moreover, the sample mean of $y_i^2(t)$ is often very similar to the first FPC. As a result, choosing $M < 4$ in the QML setting would leave only one genuinely additional direction beyond the fixed components, making the effective dimension roughly comparable to the case $M = 1$ in the LS setting. As in Section 3.2, I use innovation functions with heavy-tailed increments and again consider $\nu \in \{3, 5, 25\}$. It should be noted that for $\nu = 3$ the result in Theorem 2.2

does not hold when the estimated FPCs are used rather than the true underlying basis functions. However, this should be less of a concern for this simulation since the finite-dimensional basis specification is violated for this simulation. The training sample size was set to $N = 500$, and I ran $Q = 100$ Monte Carlo simulations for each combination of the parameters. Due to the lack of computational resources only this smaller set of possible parameter combinations was considered.

Since the true and estimated functional parameters are no longer directly comparable through coefficient-wise mean bias and standard deviation, as in Section 3.1, I instead evaluate estimation accuracy at the functional level. I follow the approach by Cerovecki et al. and define the relative root mean squared deviations for δ and α by

$$\frac{1}{Q^{1/2}\|\delta\|} \left(\sum_{q=1}^Q \|\hat{\delta}^{(q)} - \delta\|^2 \right)^{1/2}, \quad \frac{1}{Q^{1/2}\|\alpha\|} \left(\sum_{q=1}^Q \|\hat{\alpha}^{(q)} - \alpha\|^2 \right)^{1/2}. \quad (63)$$

Here $\hat{\delta}^{(q)}$ denotes the estimate of δ obtained in the q -th Monte Carlo replication. Here, $\|\cdot\|$ denotes the L^2 norm induced by the inner product for functions as δ as well as the operator norm for operators as α and β . In practice, the operator norm is approximated numerically from the discretized matrix representation of $\hat{\alpha}^{(q)} - \alpha$, using its spectral norm. The corresponding quantity for β is defined analogously.

One of the difficulties in using orthonormal instead of non-negative, linear independent basis function arises in the numerical optimization. If the parameter vector $\hat{\theta}_n = \begin{pmatrix} \hat{\mathbf{d}} & \hat{\mathbf{A}} & \hat{\mathbf{B}} \end{pmatrix}$ is estimated as described in Section 2.3, the reconstructed discretized kernels $\hat{\alpha}$ and $\hat{\beta}$ kernels need not be pointwise non-negative unless this is imposed explicitly. The difficulty is that the constraints do not apply directly to the coefficient vector $\hat{\theta}_n$, but to the functional objects reconstructed from it. This requires a tailored numerical strategy. As a consequence, I made use of the "Cobyla" algorithm (Powell, 1994) from the "NLOpt" package (Johnson, 2007). The algorithm is a derivative-free optimization algorithm that allows the specification of equality and inequality constraints. I use a derivative-free optimization routine rather than a gradient-based method, since evaluating the gradient and Jacobian of the constraint functions would require either numerical approximation or explicit differentiation through the full recursion over the sample. I found that the gradient-free version performed similarly well while being less computationally expensive as an algorithm that utilized the gradient and Jacobian. Moreover, on an equidistant

grid with 100 points, imposing positivity of $\hat{\delta}$ and pointwise non-negativity of $\hat{\alpha}$ and $\hat{\beta}$ leads to 20,100 constraints, which makes the constrained optimization computationally demanding. As a practical compromise, I impose the constraints only on a reduced grid and then truncate any remaining negative values of the estimated functional parameters to zero after estimation. This should be understood as a computational heuristic rather than as part of the estimator itself. In my simulations, this procedure reduced computation time substantially, while on average fewer than 30 grid values had to be truncated to zero. This reduced grid-projection was only used for LS estimation since the usage of non-negative baseline function does not require this procedure.

Table 2: Relative mean squared deviation for LS and QML estimation

M	$\nu = 3$			$\nu = 5$			$\nu = 25$		
	$\hat{\delta}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\delta}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\delta}$	$\hat{\alpha}$	$\hat{\beta}$
LS Estimation									
2	.4422	1.5258	2.2446	.4200	1.3762	1.9983	.4186	1.3044	1.7588
4	.4222	1.6589	2.3501	.4322	1.7076	2.1199	.4312	1.6908	1.8820
QML Estimation									
4	.7455	1.2722	1.8392	.7354	1.0149	1.8164	.7073	.9045	1.7947
6	.8286	1.4056	2.1909	.8550	1.2299	2.1698	.9703	.9865	2.1582

Table 2 reports the relative root mean squared deviations for LS and QML estimation. In this setting, the comparison between the two estimators is less clear-cut than in Section 3.1. While the deviation for $\hat{\delta}$ is generally lower for LS estimation, it tends to be higher for $\hat{\alpha}$. For $\hat{\beta}$, the differences between the two methods are smaller and do not yield a completely uniform ranking. Similarly to the results in Section 3.1, heavier tails, that is, lower values of ν , tend to be associated with larger deviations. This contrast is particularly visible when comparing $\nu = 3$ and $\nu = 25$, especially for the estimation of α and β , although the pattern is not perfectly monotone in every entry. In addition, increasing the number of retained baseline functions does not improve performance for LS estimation and appears to worsen performance for QML estimation in this simulation. One possible explanation is that, for larger values of M , the QML estimator overfits the training sample and therefore generalizes less well to the evaluation sample. However, when comparing the root integrated squared error on the training and evaluation samples, I found no clear evidence of such behaviour. Another potential explanation for this behavior is that since QML estimation uses linearly independent baseline functions rather than orthonor-

mal basis functions numerical optimization can get unstable if the baseline functions are highly correlated. When examining individual simulations, I observed some evidence of this. LS estimation does not suffer from the same issue due to the orthonormality of the basis functions. Nevertheless, even for LS estimation, using more basis functions does not improve performance in this simulation and, if anything, leads to slightly worse results.

In the following, I compare the p-values obtained under LS and QML estimation in order to further assess their relative performance in forecasting intraday VaR. Table 3 displays

Table 3: Mean p-value and standard deviation for LS and QML estimation

M	$\nu = 3$			$\nu = 5$			$\nu = 25$		
	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$
LS Estimation									
2	.3248 (.2568)	.2940 (.2221)	.2854 (.2459)	.3289 (.2851)	.3199 (.2547)	.2671 (.2321)	.2825 (.2491)	.2833 (.2286)	.2798 (.1924)
4	.3388 (.2408)	.3193 (.2203)	.3413 (.2539)	.2893 (.2311)	.3296 (.2431)	.3072 (.2556)	.3419 (.2718)	.3181 (.2222)	.3072 (.2147)
QML Estimation									
4	.3156 (.2377)	.2890 (.2353)	.2619 (.2298)	.3129 (.2579)	.3056 (.2512)	.2519 (.2117)	.2930 (.2622)	.3053 (.2558)	.2896 (.2092)
6	.3497 (.2212)	.3168 (.2342)	.2954 (.2468)	.3174 (.2347)	.3356 (.2405)	.3069 (.2356)	.3321 (.2495)	.2967 (.2248)	.2761 (.2148)

the mean and standard deviation of the p-values for LS and QML estimation. Overall, LS and QML appear to perform similarly, and neither method dominates uniformly across all simulation settings. Interestingly, using more basis functions does not appear to reduce the mean p-value for QML estimation. This is somewhat surprising given the increase in relative root mean squared deviations observed in Table 2. In conclusion, the evidence does not support a clear overall ranking of the two estimation procedures.

A slightly clearer pattern emerges across VaR levels: more extreme quantiles, that is, smaller values of τ , tend to be associated with lower mean p-values. By contrast, the effect of the tail thickness of the innovation process is less clear, as the relationship between ν and the mean p-value is not monotone across all simulation settings. In particular, the inclusion of the case $\nu = 3$ does not produce a uniformly worse p-value profile, even though heavier tails were associated with larger estimation errors in Table 2.

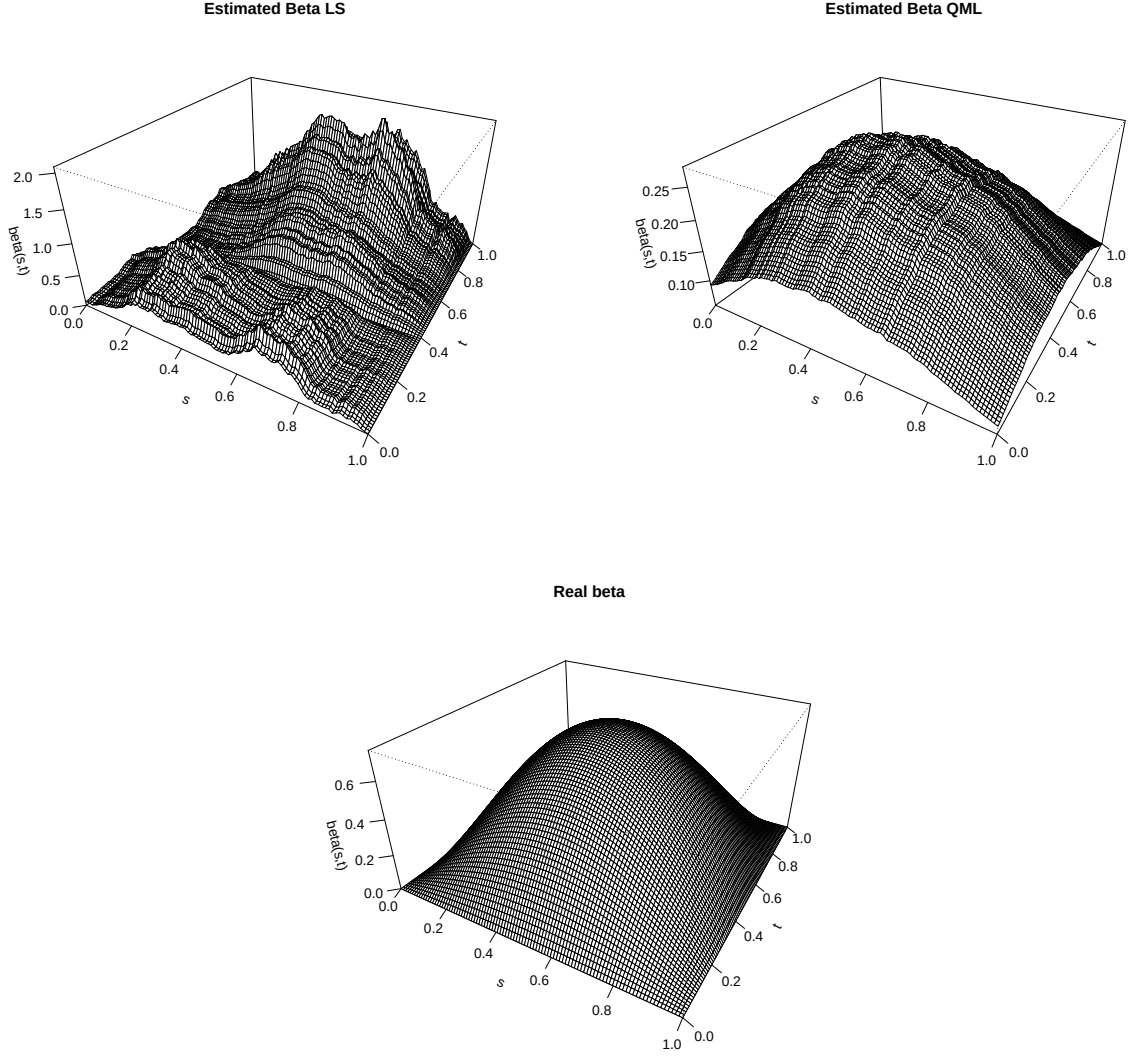


Figure 1: Illustration of one realization of the estimated kernel functions $\hat{\beta}$ under LS and QML estimation, together with the true kernel function β . The LS estimator is based on four basis functions, whereas the QML estimator uses two baseline functions. The innovation parameter is fixed at $\nu = 25$.

Since discretized functional data can be represented as high-dimensional multivariate observations, it is natural to consider a multivariate GARCH model for forecasting intraday VaR. Because observations at nearby grid points are typically highly correlated, a PC-GARCH model appears to be a plausible specification for such data. Consequently, I compared the finite sample performance of the PC-GARCH model under the current simulation specification to the results of the FGARCH(1, 1) model.

Remark 3.1: Comparison between the discretized FGARCH(1,1) model and the PC-GARCH model

As discussed previously, functional data are observed in practice on a finite grid and can therefore be represented in multivariate form. More precisely, a discretized version of the FGARCH(1, 1) model can be written as

$$y_i = \sigma_i \odot \varepsilon_i \tag{64}$$

$$\sigma_i^2 = \delta + \alpha y_{i-1}^2 + \beta \sigma_{i-1}^2 \tag{65}$$

where $\odot$ denotes the Hadamard product. Moreover, y_i , σ_i , ε_i , σ_i^2 , y_i^2 , and δ are vectors of length T , where T is the number of grid points, while α and β are $T \times T$ matrices. Here, σ_i^2 and y_i^2 denote the pointwise squares of σ_i and y_i , that is, $\sigma_i^2 = \sigma_i \odot \sigma_i$ and $y_i^2 = y_i \odot y_i$.

If, in addition, the basis functions in the finite-dimensional representation of the FGARCH model are estimated by FPCA, as was done for the LS estimator in Section 3.2, then both the PC-GARCH and FGARCH models involve a reduction of dimensionality through a projection onto PCs. In this sense, the two approaches are similar in practice. Nevertheless, even under this specific setup, there remain important differences between them. First, while the PC-GARCH model models the full conditional covariance matrix of y_i by assuming that it is spanned by a fixed set of PCs, the FGARCH model, under this finite-dimensional specification, is driven by a recursion for projected quantities associated with y_i^2 and σ_i^2 , which are then combined with the basis functions to recover the pointwise conditional variance. Second, the PCs entering the PC-GARCH model are typically estimated from the covariance structure of y_i , whereas in the FGARCH framework they are obtained from the covariance structure of y_i^2 . Third, while the FGARCH model allows dynamic interaction across components, the PC-GARCH model assumes that each latent variance component

evolves through its own univariate GARCH recursion. Consequently, intuitively, the FGARCH model allows for cross-component dependence already at the level of the projected variance dynamics, while the PC-GARCH model keeps the latent variance dynamics separate and introduces dependence only through the reconstruction of the covariance matrix along fixed principal-component directions.

Table 4: Mean p-value and standard deviation for PC-GARCH

Number PCs used	$\nu = 3$			$\nu = 5$			$\nu = 25$		
	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$
2	.2826 (.2796)	.3845 (.2870)	.3313 (.2797)	.1581 (.2078)	.2816 (.2733)	.3062 (.3098)	.0728 (.1381)	.1475 (.2308)	.2034 (.2869)
4	.0062 (.0181)	.1008 (.1995)	.1518 (.2451)	.0102 (.0399)	.0840 (.1926)	.1083 (.2142)	.0002 (.0014)	.0229 (.0698)	.0615 (.1812)

The results in Table 4 indicate that the PC-GARCH model remains a reasonably competitive benchmark, even though the data-generating process is an FGARCH(1, 1) model. The use of more PCs generally results in lower mean p-values. The corresponding reduction in standard deviation is likely related to the fact that the p-values become concentrated near the lower boundary of the unit interval, which mechanically reduces their dispersion. Thus, standard deviations are especially low when mean p-values are also low. To better understand why using more PCs leads to worse performance, I compared the fitted innovation functions implied by the PC-GARCH model with the true innovation functions. This comparison suggests that, for larger numbers of PCs, the PC-GARCH model tends to overfit the training data. In addition, the PC-GARCH model appears to perform better for lower values of ν , as the mean p-values are generally higher when the innovation process is more heavy-tailed. This pattern contrasts with the FGARCH-based estimators, for which the effect of ν on the p-values was less clear-cut. Nevertheless, these results suggest that PC-GARCH can be a meaningful benchmark for FGARCH(1, 1), even when the underlying data-generating process is functional.

4 Discussion

In this thesis, the FGARCH model was reviewed and sufficient conditions for the existence of a strictly stationary and non-anticipative solution were summarized. In addition,

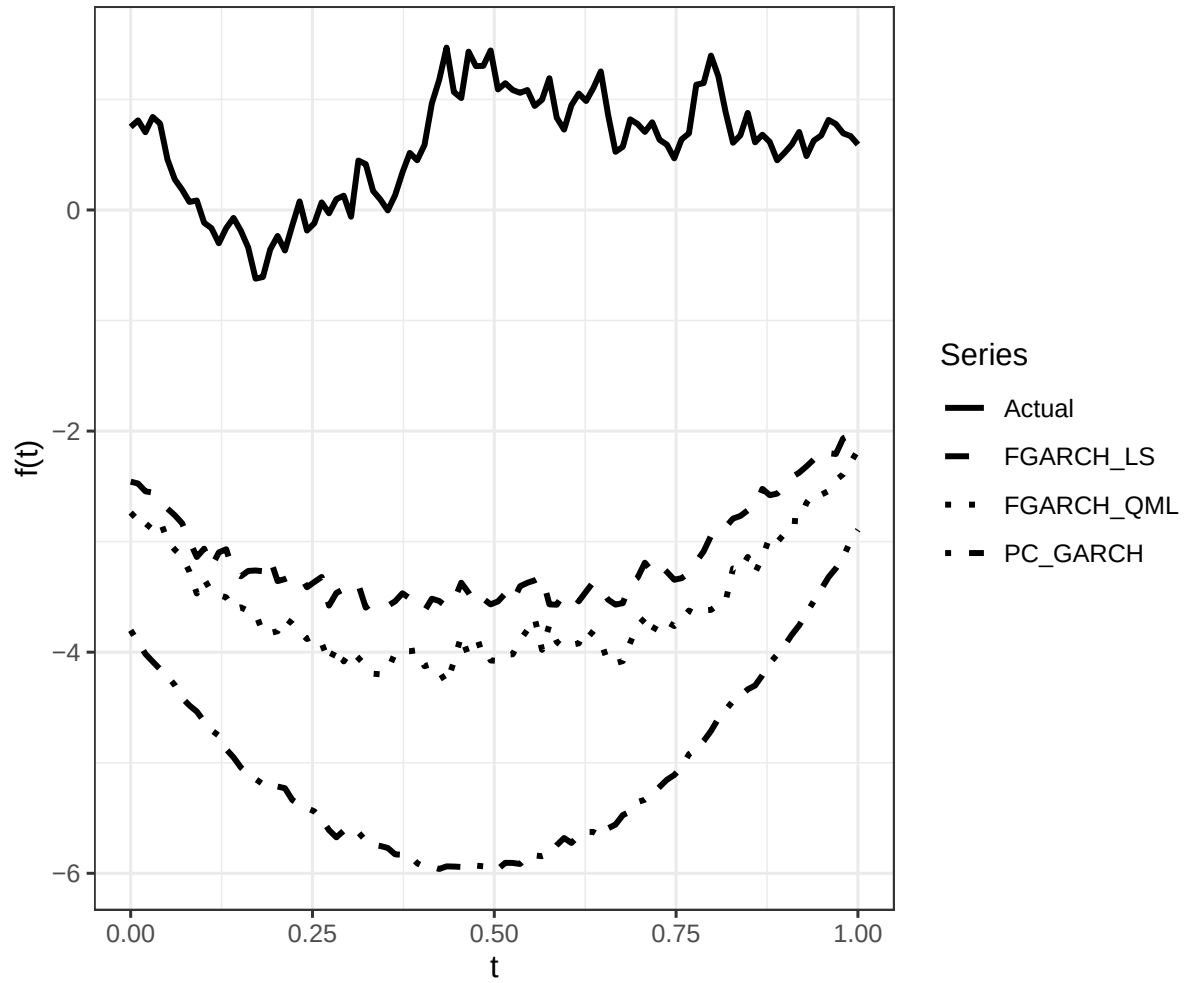


Figure 2: Illustration of one realization of the one-day-ahead intraday VaR forecasts under FGARCH LS estimation, FGARCH QML estimation, and PC-GARCH, together with the realized series. The LS estimator is based on four basis functions, the QML estimator uses two baseline functions and the PC-GARCH model uses two PCs. The innovation parameter is fixed at $\nu = 25$.

conditions for the existence of moments and for L^p - m -approximability were discussed, together with the difficulty of establishing necessary conditions for stationarity in the infinite-dimensional setting due to the non-equivalence of norms. A further limitation is that stronger mixing conditions are difficult to establish for the FGARCH model, mainly because the infinite-dimensional setting generally rules out density-based arguments that are available in finite-dimensional frameworks. The thesis also reviewed two estimation procedures for the FGARCH model and investigated their finite-sample performance by means of Monte Carlo simulations. Both procedures rely on a dimension-reduction assumption to a fixed finite dimension, which constitutes an important limitation. Although Cerovecki et al. extend the QML approach to infinite-dimensional parameter spaces and establish consistency in that setting, they also note that deriving an asymptotic distribution remains difficult. On the empirical side, the Monte Carlo study shows that, in a simple simulation setting in which the true underlying basis function is assumed to be known and only one basis function is used, QML estimation outperforms LS estimation. The results further indicate that thinner-tailed innovation increments and larger sample sizes lead to more accurate parameter estimates and improved intraday VaR forecasts. When the true underlying basis functions are not known and the assumptions underlying the estimation procedures are violated, however, the comparison becomes less clear. In particular, both methods perform similarly when the assumption that all functional parameters can be represented using a common set of basis functions is violated. Overall, these findings suggest that the relative performance of the two estimation procedures depends strongly on how closely the simulation setting matches their underlying assumptions. Finally, the results show that the PC-GARCH model performs relatively well compared with the FGARCH model even when the data are generated from an FGARCH process. Practical implementation issues were also highlighted, especially for the LS estimator when orthonormal eigenfunctions are used as basis functions.

The scope of this thesis is deliberately limited. It does not aim to develop new theoretical results for the FGARCH model, nor does it attempt to provide a comprehensive comparison with all competing approaches to multivariate or functional volatility modelling. Instead, the thesis focuses on presenting and contextualizing the existing theory and on studying the finite-sample behaviour of the FGARCH model and its estimation procedures in a Monte Carlo setting with a focus on intraday VaR forecasting. Likewise, although the limitations of VaR are discussed, the thesis does not seek to provide a

broader comparative assessment of alternative risk measures.

There are several directions in which the analysis in this thesis could be extended. One possible extension concerns alternative estimation procedures. Kühnert (2020) proposes an alternative estimator for the FGARCH model based on Yule-Walker equations. The author also derives explicit asymptotic upper error bounds of estimation errors on finite-dimensional subspaces. His estimation procedure allows the functional operators to be estimated more directly, without relying on the finite-dimensional parametrizations used for the QML and LS estimators in this thesis. A detailed discussion of this estimation procedure was beyond the scope of this thesis.

Another direction in which FGARCH models can be further developed is through alternative model specifications. Sun and Yu (2020) proposes a functional threshold GARCH model that is designed to capture asymmetric responses of the volatility process to unanticipated shocks. Rice et al. (2023) introduces a FGARCH model with an additional exogenous covariate function intended to capture long-range conditional heteroskedasticity. They use the model to forecast the intraday return curves of crude oil contracts. Li et al. (2025) establishes an FGARCH model with multiple constant parameters and present an estimation procedure for it. The addition of multiple constant parameters extends the dimension of the space in which the conditional volatility curves are located in. They find that the FGARCH model with multiple constant parameters outperforms the standard FGARCH model in a real data application in both in-sample fitting and out-of-sample forecasting. Furthermore, in a recent contribution, Aue et al. (2026) introduces an operator-level ARCH model that is meant to model the full evolution of the conditional covariance operator rather than just the pointwise variance. To the best of my knowledge, the incorporation of multiple functional processes into a genuinely multivariate FGARCH framework remains unexplored.

A further extension concerns inferential tools for FGARCH models. This thesis also does not address the issue of testing for conditional heteroskedasticity and selection of appropriate orders (p, q) for an FGARCH model. Rice et al. (2020b) studies these issues and propose statistical procedures for order selection and for testing the assumption of conditional heteroskedasticity.

Another possible extension concerns the choice of basis and baseline functions used in the

LS and QML estimation procedures. In Rice et al. (2023), the authors discuss alternative methods for estimating the non-negative baseline functions used in the QML estimation of both the FGARCH model and their FGARCH-X extension. They propose a sparse, non-negative version of FPCA as well as the use of truncated predictive factors based on the estimation procedure from Kargin and Onatski (2008). Several other specifications of basis functions could be considered. One noteworthy alternative, which to my knowledge has not yet been explored in the context of FGARCH models, is the use of smoothed FPCs introduced by Silverman (1996). The author shows that by replacing the usual L^2 -orthonormality of the eigenfunctions by orthonormality with respect to an inner product that takes account of the roughness of the functions the estimation procedure effectively reduces the noise contained in the FPCs. This could be useful in the context of intraday return data which is known to exhibit microstructure noise at sufficiently fine sampling frequencies (Cerovecki et al., 2019).

Finally, the estimation of VaR for intraday return curves is also explored in other practical applications. Kearney et al. (2025) uses the FGARCH model to forecast VaR for intraday return curves of foreign exchange currency data for four major currencies. They also discuss two additional basis specifications, which further illustrates the practical importance of basis choice in FGARCH applications.

Forecasting intraday return curves remains a promising and practically relevant area of research. While the FGARCH framework provides a flexible and theoretically appealing approach, important challenges remain. At the same time, these limitations point to several worthwhile directions for further methodological and empirical work. For this reason, the topic appears to offer considerable potential for future research and may be one that I would like to examine further beyond the scope of this thesis.

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A Appendix

A.1 Availability of the Simulation Codes

The simulation codes can be found in the following GitHub repository: https://github.com/schmidtfabian/functional_garch.

A.2 Results of Section 2.1

A.2.1 Definition of Ergodicity

This definition is taken from Francq and Zakoian (2019).

Definition A.1 (Ergodic stationary process). A real valued strictly stationary process $(Z_i)_{i \in \mathbb{Z}}$ is said to be ergodic if and only if, for any Borel set B and any integer k ,

$$\frac{1}{n} \sum_{i=1}^n \mathbb{1}_B(Z_i, Z_{i+1}, \dots, Z_{i+k}) \rightarrow \mathbb{P}[(Z_1, \dots, Z_{1+k}) \in B] \quad (66)$$

with probability 1.

It should be noted that the concept of ergodicity is much more general and can be extended to non-stationary sequences (see, for example, Billingsley (1995)).

The ergodic theorem applied to stationary sequences states the following.

Theorem A.1 (The ergodic theorem for stationary sequences). If $(Z_i)_{i \in \mathbb{Z}}$ is strictly stationary and ergodic, if f is measurable and if

$$\mathbb{E}|f(\dots, Z_{i-1}, Z_i, Z_{i+1}, \dots)| < \infty \quad (67)$$

then

$$\frac{1}{n} \sum_{i=1}^n f(\dots, Z_{i-1}, Z_i, Z_{i+1}, \dots) \rightarrow \mathbb{E}[f(\dots, Z_{i-1}, Z_i, Z_{i+1}, \dots)] \quad a.s. \quad (68)$$

A.2.2 Proof of Theorem 2.1

As noted in equation (5) the conditional volatility equation of the scalar GARCH(1, 1) model can be rewritten into

$$\sigma_t^2 = \delta + \gamma_{t-1} \sigma_{t-1}^2. \quad (69)$$

Iterating yields

$$\sigma_t^2 = \delta \left(1 + \sum_{i=1}^J \overbrace{\gamma_{t-1} \dots \gamma_{t-i}}^{\Gamma_{t,i}} \right) + \overbrace{\gamma_{t-1} \dots \gamma_{t-J-1}}^{\Gamma_{t,J}} \sigma_{t-J-1}^2 \quad (70)$$

$$:= h_t(J) + \overbrace{\gamma_{t-1} \dots \gamma_{t-J-1}}^{\Gamma_{t,J}} \sigma_{t-J-1}^2 \quad (71)$$

Letting $J \rightarrow \infty$ identifies h_t as a potential solution candidate

$$h_t := \delta \left(1 + \sum_{j=1}^{\infty} \Gamma_{t,j} \right), \quad \text{where } \Gamma_{t,i} = \gamma_{t-1} \dots \gamma_{t-i}. \quad (72)$$

To show that h_i converges, I make use of the Cauchy Root Test for non-negative terms.

Definition A.2 (Cauchy Root Test). Let $\sum a_n$ be a series with non-negative terms and let $\lambda = \overline{\lim} a_n^{1/n}$. Then (i) if $\lambda < 1$ the series $\sum a_n$ converges, (ii) if $\lambda > 1$ the series $\sum a_n$ diverges.

By the Cauchy Root Test, $\Gamma_{t,i}$ has to satisfy the following equation

$$\lambda := [\Gamma_{t,i}]^{1/i} < 1. \quad (73)$$

To show that this inequality holds, rewrite

$$\lambda = [\Gamma_{t,J}]^{1/J} = \exp \left[\frac{1}{J} \log (\Gamma_{t,J}) \right] = \exp \left[\frac{1}{J} \log (\gamma_{t-1} \dots \gamma_{t-J}) \right] = \exp \left[\frac{1}{J} \sum_{i=1}^J \log (\gamma_{t-i}) \right]. \quad (74)$$

By an application of the law of large numbers $\frac{1}{J} \sum_{i=1}^J \log (\gamma_{t-i})$ converges to $\mathbb{E}[\log \{\overbrace{\alpha \varepsilon_t^2}^{\gamma_t} + \beta\}]$ and $\lambda < 1$. Consequently, h_t converges.

As a second step, we have to prove that h_t is the unique solution to the scalar GARCH(1, 1)

model. This proof is done by contradiction. To do so, assume that $\tilde{\sigma}_t^2$ denotes an alternative solution to the scalar GARCH(1, 1) model. By equation (70), $\tilde{\sigma}_t$ has to satisfy

$$\sigma_t^2 = h_t(J) + \gamma_{t-1} \dots \gamma_{t-J-1} \tilde{\sigma}_{t-J-1}^2. \quad (75)$$

For $\tilde{\sigma}_t$ to be an alternative solution to h_t their difference has to be greater than 0

$$\tilde{\sigma}_t - h_t = (h_t(J) - h_t) + \gamma_{t-1} \dots \gamma_{t-J-1} \tilde{\sigma}_{t-J-1}^2. \quad (76)$$

As $J \rightarrow \infty$, the right-hand side of the equation above goes to 0. Thus, h_t and $\tilde{\sigma}_t$ coincide. Therefore, h_t is the unique solution of the scalar GARCH(1, 1) model. That the solution is non-anticipative follows from equation (72). Furthermore, equation (72) implies that there exists a functional g such that

$$\sigma_t^2 = g(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots), \quad t \in \mathbb{Z} \quad (77)$$

which implies that σ_t^2 are Bernoulli shifts. The ergodicity of the solution is then implied by the work of Stout (1974). $\square$

A.3 Proofs of Section 2.2

A.3.1 Proof of Theorem 2.2

As stated previously, the proof of Theorem 2.2 follows analogous arguments to the proof of Theorem 2.1. First, rewrite the volatility equation in the random coefficient form

$$\sigma_i^2(t) = \delta(t) + \int \gamma_{i-1}(t, s) \sigma_{i-1}^2(s) ds. \quad (78)$$

Iterating backwards yields

$$\begin{aligned}
\sigma_i^2(t) &= \delta(t) + \int \gamma_{i-1}(t, s_1) \sigma_{i-1}^2(s_1) ds_1 \\
&= \delta(t) + \int \gamma_{i-1}(t, s_1) \delta(s_1) ds_1 + \iint \gamma_{i-1}(t, s_1) \gamma_{i-2}(s_1, s_2) \sigma_{i-2}^2(s_2) ds_2 ds_1 \\
&= \delta(t) + \int \gamma_{i-1}(t, s_1) \delta(s_1) ds_1 + \iint \gamma_{i-1}(t, s_1) \gamma_{i-2}(s_1, s_2) \delta(s_2) ds_2 ds_1 \\
&\quad + \iiint \gamma_{i-1}(t, s_1) \gamma_{i-2}(s_1, s_2) \gamma_{i-3}(s_2, s_3) \sigma_{i-3}^2(s_3) ds_3 ds_2 ds_1,
\end{aligned}$$

and so on.

For $i \in \mathbb{Z}$, let $\Gamma_{i,0}$ denote the identity operator, and for $k \in \mathbb{N}$ define

$$(\Gamma_{i,k}x)(t) = \int \cdots \int \gamma_{i-1}(t, s_1) \gamma_{i-2}(s_1, s_2) \cdots \gamma_{i-k}(s_{k-1}, s_k) x(s_k) ds_k \cdots ds_1. \quad (79)$$

Then, for every $m \in \mathbb{N}$,

$$\sigma_i^2 = \sum_{k=0}^{m-1} \Gamma_{i,k} \delta + \Gamma_{i,m} \sigma_{i-m}^2. \quad (80)$$

Moreover,

$$\Gamma_{i,k} = \Gamma_{i,1} \circ \Gamma_{i-1,1} \circ \cdots \circ \Gamma_{i-k+1,1} \quad (81)$$

where $\circ$ denotes the composition of operators. This identifies

$$\sigma_i^2 = \sum_{k=0}^{\infty} \Gamma_{i,k} \delta \quad (82)$$

as a candidate for a nonanticipative solution. It remains to show that this series is well defined almost surely in L^2 .

Since $\Gamma_{i,1}$ is the integral operator with kernel γ_{i-1} , we have

$$\|\Gamma_{i,1}\|_{\mathcal{N}} \leq \|\gamma_{i-1}\|_S. \quad (83)$$

This follows by an application of the Cauchy-Schwarz inequality (Riesz and Nagy, 1990). To justify this bound, note that for each $j \in \mathbb{Z}$, the operator $\Gamma_{j,1}$ is the integral operator

on L^2 with kernel γ_{j-1} , that is,

$$(\Gamma_{j,1}x)(t) = \int \gamma_{j-1}(t, s)x(s) ds.$$

Hence, by the Cauchy–Schwarz inequality,

$$|(\Gamma_{j,1}x)(t)|^2 = \left| \int \gamma_{j-1}(t, s)x(s) ds \right|^2 \leq \left(\int \gamma_{j-1}(t, s)^2 ds \right) \left(\int x(s)^2 ds \right).$$

Integrating with respect to t gives

$$\|\Gamma_{j,1}x\|_2^2 \leq \left(\iint \gamma_{j-1}(t, s)^2 ds dt \right) \|x\|_2^2 = \|\gamma_{j-1}\|_S^2 \|x\|_2^2.$$

Therefore,

$$\|\Gamma_{j,1}\|_{\mathcal{N}} \leq \|\gamma_{j-1}\|_S.$$

Using equation (81) and the submultiplicativity of the operator norm, it follows that

$$\|\Gamma_{i,k}\|_{\mathcal{N}} \leq \prod_{r=0}^{k-1} \|\Gamma_{i-r,1}\|_{\mathcal{N}} \leq \prod_{\ell=1}^k \|\gamma_{i-\ell}\|_S.$$

Hence, by equation (81),

$$\|\Gamma_{i,k}\|_{\mathcal{N}} \leq \prod_{\ell=1}^k \|\gamma_{i-\ell}\|_S. \quad (84)$$

Because the random kernels are i.i.d. and $E \log \|\gamma_0\|_S < 0$, the strong law of large numbers implies

$$\frac{1}{k} \sum_{\ell=1}^k \log \|\gamma_{i-\ell}\|_S \rightarrow E \log \|\gamma_0\|_S < 0 \quad \text{a.s.} \quad (85)$$

Therefore,

$$\sum_{k=0}^{\infty} \|\Gamma_{i,k}\delta\|_2 \leq \|\delta\|_2 \sum_{k=0}^{\infty} \|\Gamma_{i,k}\|_{\mathcal{N}} < \infty \quad \text{a.s.} \quad (86)$$

It follows that $\sum_{k=0}^{\infty} \Gamma_{i,k}\delta$ converges absolutely in L^2 almost surely.

Next, using $\Gamma_{i,k+1} = \Gamma_{i,1} \circ \Gamma_{i-1,k}$, we obtain

$$\begin{aligned} \sum_{k=0}^{\infty} \Gamma_{i,k} \delta &= \delta + \sum_{k=1}^{\infty} \Gamma_{i,k} \delta \\ &= \delta + \Gamma_{i,1} \sum_{k=0}^{\infty} \Gamma_{i-1,k} \delta. \end{aligned}$$

Hence the series indeed satisfies the volatility recursion, so it defines a nonanticipative solution.

To prove uniqueness, let $(\tilde{\sigma}_i^2)$ be another strictly stationary nonanticipative solution. Then, for every $m \in \mathbb{N}$,

$$\tilde{\sigma}_i^2 = \sum_{k=0}^{m-1} \Gamma_{i,k} \delta + \Gamma_{i,m} \tilde{\sigma}_{i-m}^2. \quad (87)$$

Therefore,

$$\begin{aligned} \left\| \tilde{\sigma}_i^2 - \sum_{k=0}^{\infty} \Gamma_{i,k} \delta \right\|_2 &\leq \left\| \sum_{k=m}^{\infty} \Gamma_{i,k} \delta \right\|_2 + \|\Gamma_{i,m} \tilde{\sigma}_{i-m}^2\|_2 \\ &\leq \sum_{k=m}^{\infty} \|\Gamma_{i,k} \delta\|_2 + \|\Gamma_{i,m}\|_{\mathcal{N}} \|\tilde{\sigma}_{i-m}^2\|_2. \end{aligned}$$

The first term tends to 0 almost surely. For the second term, $\|\Gamma_{i,m}\|_{\mathcal{N}} \rightarrow 0$ almost surely, and since $(\tilde{\sigma}_i^2)$ is strictly stationary, the sequence $(\|\tilde{\sigma}_{i-m}^2\|_2)_m$ is bounded in probability. Hence

$$\|\Gamma_{i,m}\|_{\mathcal{N}} \|\tilde{\sigma}_{i-m}^2\|_2 \xrightarrow{p} 0. \quad (88)$$

Thus, for every $\varepsilon > 0$,

$$\mathbb{P} \left(\left\| \tilde{\sigma}_i^2 - \sum_{k=0}^{\infty} \Gamma_{i,k} \delta \right\|_2 > \varepsilon \right) = 0, \quad (89)$$

which implies

$$\tilde{\sigma}_i^2 = \sum_{k=0}^{\infty} \Gamma_{i,k} \delta \quad \text{a.s.} \quad (90)$$

Therefore, the strictly stationary nonanticipative solution is unique.

Next, we are proving part (ii) from Theorem 2.2. Assume that there exists $\nu > 0$ such that

$$E\|\gamma_0\|_S^\nu < 1.$$

First note that, by Jensen's inequality,

$$E \log \|\gamma_0\|_S = \frac{1}{\nu} E \log \|\gamma_0\|_S^\nu \leq \frac{1}{\nu} \log E\|\gamma_0\|_S^\nu < 0.$$

Hence the existence and uniqueness of a strictly stationary nonanticipative solution follow from the proof of (22).

It remains to prove that $E\|\sigma_0^2\|_2^\nu < \infty$. Using the series representation

$$\sigma_0^2 = \sum_{k=0}^{\infty} \Gamma_{0,k} \delta$$

and the bound

$$\|\Gamma_{0,k}\|_{\mathcal{N}} \leq \prod_{\ell=1}^k \|\gamma_{-\ell}\|_S,$$

we obtain

$$\|\sigma_0^2\|_2 \leq \sum_{k=0}^{\infty} \|\Gamma_{0,k} \delta\|_2 \leq \|\delta\|_2 \sum_{k=0}^{\infty} \|\Gamma_{0,k}\|_{\mathcal{N}} \leq \|\delta\|_2 \sum_{k=0}^{\infty} \prod_{\ell=1}^k \|\gamma_{-\ell}\|_S.$$

If $\nu \geq 1$, Minkowski's inequality yields

$$(E\|\sigma_0^2\|_2^\nu)^{1/\nu} \leq \|\delta\|_2 \sum_{k=0}^{\infty} \left(E \left[\prod_{\ell=1}^k \|\gamma_{-\ell}\|_S^\nu \right] \right)^{1/\nu}.$$

Since the γ_j are i.i.d., this becomes

$$(E\|\sigma_0^2\|_2^\nu)^{1/\nu} \leq \|\delta\|_2 \sum_{k=0}^{\infty} (E\|\gamma_0\|_S^\nu)^{k/\nu},$$

and the right-hand side is finite because $E\|\gamma_0\|_S^\nu < 1$.

If $0 < \nu \leq 1$, then the function $x \mapsto x^\nu$ is subadditive on $[0, \infty)$, so

$$\|\sigma_0^2\|_2^\nu \leq \|\delta\|_2^\nu \sum_{k=0}^{\infty} \left(\prod_{\ell=1}^k \|\gamma_{-\ell}\|_S \right)^\nu.$$

Taking expectations and again using independence gives

$$E\|\sigma_0^2\|_2^\nu \leq \|\delta\|_2^\nu \sum_{k=0}^{\infty} E \left[\prod_{\ell=1}^k \|\gamma_{-\ell}\|_S^\nu \right] = \|\delta\|_2^\nu \sum_{k=0}^{\infty} (E\|\gamma_0\|_S^\nu)^k < \infty.$$

Therefore, in both cases,

$$E\|\sigma_0^2\|_2^\nu < \infty.$$

This completes the proof. □

A.4 Proofs of Section 2.3

A.4.1 Proof of Theorem 2.4

To establish the asymptotic distribution of the LS estimator, we start by making use of the component-wise mean value theorem. Using the mean value theorem to prove the consistency or asymptotic distribution of an estimator is not uncommon among m-estimators. Likewise, the consistency and asymptotic distribution of maximum likelihood estimator is proven similarly. The proof of Theorem 2.4 relies on a component-wise application of the mean value theorem. Define

$$\hat{Z}_n(\underline{\theta}) = \frac{1}{n} \sum_{i=2}^n \left\{ y_i^{(2)} - \hat{s}_i^{(2)}(\underline{\theta}) \right\}^\top \left\{ y_i^{(2)} - \hat{s}_i^{(2)}(\underline{\theta}) \right\},$$

the sample LS criterion for a generic parameter vector $\underline{\theta}$. Denote its gradient and Hessian by $\frac{\partial \hat{Z}_n(\underline{\theta})}{\partial \underline{\theta}}$ and $\frac{\partial^2 \hat{Z}_n(\underline{\theta})}{\partial \underline{\theta}^2}$, respectively. Applying the mean value theorem component-wise to the gradient of $\hat{Z}_n$ between the true parameter θ and the estimator $\hat{\underline{\theta}}_n$ yields

$$\frac{\partial \hat{Z}_n(\hat{\underline{\theta}}_n)}{\partial \underline{\theta}} - \frac{\partial \hat{Z}_n(\theta)}{\partial \underline{\theta}} = \frac{\partial^2 \hat{Z}_n(\check{\theta})}{\partial \underline{\theta}^2} (\hat{\underline{\theta}}_n - \theta),$$

for some intermediate value $\check{\theta}$ lying between θ and $\hat{\underline{\theta}}_n$. Consequently, to derive the asymptotic distribution of $\hat{\underline{\theta}}_n$ we need to derive the unknown quantities in the mean value theorem. To do so, first note that

$$\frac{\partial \hat{Z}_n(\hat{\underline{\theta}}_n)}{\partial \underline{\theta}} = 0 \tag{91}$$

by construction since $\hat{\underline{\theta}}_n$ minimizes $\hat{Z}_n$.

Next, the gradient and Hessian of $\hat{Z}_n$ with respect to the true parameter vector θ is equal to

$$\frac{\partial \tilde{Z}_n(\theta)}{\partial \theta} = -\frac{2}{n} \sum_{i=2}^n \left(\frac{\partial \tilde{s}_i^{(2)}(\theta)}{\partial \theta} \right)^\top \{y_i^{(2)} - \tilde{s}_i^{(2)}(\theta)\}$$

and

$$\frac{\partial^2 \tilde{Z}_n(\theta)}{\partial \theta^2} = \frac{2}{n} \sum_{i=2}^n \left(\frac{\partial \tilde{s}_i^{(2)}(\theta)}{\partial \theta} \right)^\top \left(\frac{\partial \tilde{s}_i^{(2)}(\theta)}{\partial \theta} \right) + \frac{2}{n} \sum_{i=2}^n R_i(\theta),$$

where

$$R_i(\theta) = \left\{ \sum_{m=1}^M \left(\frac{\partial^2 \tilde{s}_{i,m}^{(2)}(\theta)}{\partial \theta_j \partial \theta_k} \right) (\langle y_i^2, \varphi_m \rangle - \tilde{s}_{i,m}^{(2)}(\theta)) : j, k = 1, \dots, 2M^2 + M \right\},$$

with

$$\theta = (\theta_1, \dots, \theta_{2M^2+M})^\top \quad \text{and} \quad \tilde{s}_i^{(2)}(\theta) = (\tilde{s}_{i,1}^{(2)}(\theta), \dots, \tilde{s}_{i,M}^{(2)}(\theta))^\top.$$

We define

$$Q(\theta) = E \left[\left(\frac{\partial \tilde{s}_0^{(2)}(\theta)}{\partial \theta} \right)^\top \left(\frac{\partial \tilde{s}_0^{(2)}(\theta)}{\partial \theta} \right) \right] + E[R_0(\theta)].$$

It is easy to see that

$$Q(\theta) \text{ is continuous on } \Theta \quad \text{and} \quad Q_0 = Q(\theta). \quad (92)$$

Hence, the ergodic theorem (Theorem A.1) implies that

$$\sup_{\theta \in \Theta} \left\| \frac{\partial^2 \tilde{Z}_n(\theta)}{\partial \theta^2} - 2Q(\theta) \right\| \rightarrow 0 \quad \text{a.s.} \quad (93)$$

The central limit theorem for orthogonal martingale differences in Billingsley (1968) yields moreover that

$$\sqrt{n} \frac{1}{2} \frac{\partial \tilde{Z}_n(\theta)}{\partial \theta} \xrightarrow{\mathcal{D}} \tilde{N}, \quad (94)$$

where $\tilde{N}$ is a $2M^2 + M$ -dimensional normal random vector with

$$E[\tilde{N}] = 0 \quad \text{and} \quad E[\tilde{N} \tilde{N}^\top] = H_0^\top J_0 H_0.$$

Since

$$\frac{\partial \tilde{Z}_n(\theta)}{\partial \theta} = -\frac{2}{n} \sum_{i=2}^n \left(\frac{\partial \tilde{s}_i^{(2)}(\theta)}{\partial \theta} \right)^\top (y_i^{(2)} - \tilde{s}_i^{(2)}(\theta)),$$

the relevant summands are of the form

$$\left(\frac{\partial \tilde{s}_i^{(2)}(\theta)}{\partial \theta} \right)^\top (y_i^{(2)} - \tilde{s}_i^{(2)}(\theta)).$$

Here, $\partial \tilde{s}_i^{(2)}(\theta)/\partial \theta$ is determined by the past, whereas $y_i^{(2)} - \tilde{s}_i^{(2)}(\theta)$ is the new centered innovation at time i . Hence the summands form a martingale difference sequence, and the terms corresponding to different i are orthogonal. Therefore the central limit theorem for orthogonal martingale differences can be applied to obtain equation (94). By the consistency of $\hat{\theta}_n$, one can verify that

$$n \sup_{\theta \in \Theta} \left\| \frac{\partial \tilde{Z}_n(\theta)}{\partial \theta} - \frac{\partial \hat{Z}_n(\theta)}{\partial \theta} \right\| = O(1) \quad \text{a.s.} \quad (95)$$

and

$$n \sup_{\theta \in \Theta} \left\| \frac{\partial^2 \tilde{Z}_n(\theta)}{\partial \theta^2} - \frac{\partial^2 \hat{Z}_n(\theta)}{\partial \theta^2} \right\| = O(1) \quad \text{a.s.} \quad (96)$$

Therefore,

$$\frac{\partial^2 \hat{Z}_n(\check{\theta})}{\partial \theta^2} \rightarrow 2Q_0. \quad (97)$$

Combing these results the asymptotic normality of the LS estimator is established. $\square$

A.5 Additional Tables and Plots from Section 3

Table 5: Mean p-value and standard deviation for LS

Sample size	$\nu = 3$			$\nu = 5$			$\nu = 25$		
	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$
250	.2227	.2663	.2402	.2851	.2997	.2665	.3137	.2714	.2495
	(.2100)	(.2591)	(.2226)	(.2550)	(.2510)	(.2408)	(.2382)	(.1953)	(.1937)
500	.3161	.2866	.2628	.3261	.3630	.3326	.3528	.3088	.2876
	(.2644)	(.2646)	(.2773)	(.2549)	(.2541)	(.2420)	(.2488)	(.2397)	(.2305)
1000	.3674	.3942	.3599	.3784	.4071	.3427	.3726	.3557	.2785
	(.2481)	(.2697)	(.2949)	(.2494)	(.2669)	(.2551)	(.2444)	(.2319)	(.2147)

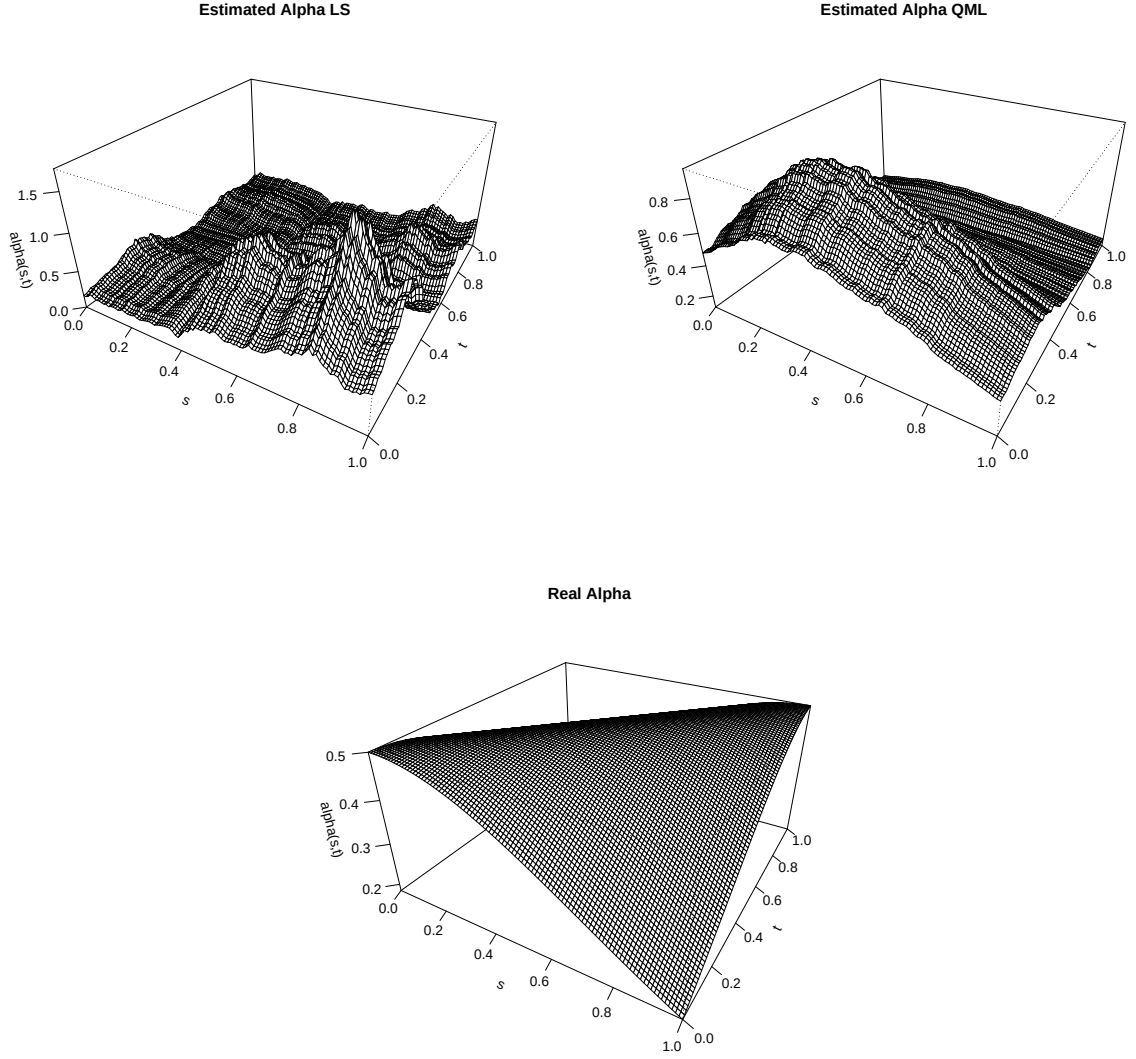


Figure 3: Illustration of one realization of the estimated kernel functions $\hat{\alpha}$ under LS and QML estimation, together with the true kernel function α . The LS estimator is based on four basis functions, whereas the QML estimator uses two baseline functions. The innovation parameter is fixed at $\nu = 25$.

Table 6: Mean p-value and standard deviation for QML

Sample size	$\nu = 3$			$\nu = 5$			$\nu = 25$		
	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$	$\tau_{0.025}$	$\tau_{0.01}$	$\tau_{0.005}$
250	.2329 (.1971)	.2647 (.2433)	.2317 (.2319)	.3071 (.2285)	.3147 (.2495)	.2782 (.2390)	.3175 (.2392)	.2926 (.2064)	.2451 (.1871)
500	.3613 (.2842)	.3384 (.2932)	.2954 (.2787)	.3349 (.2458)	.3246 (.2323)	.3255 (.2282)	.3641 (.2476)	.3324 (.2425)	.2758 (.2009)
1000	.4049 (.2322)	.4250 (.2532)	.3605 (.2674)	.4127 (.2440)	.3922 (.2524)	.3526 (.2428)	.4069 (.2720)	.3712 (.2437)	.2975 (.2038)

Declaration

I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Datum, Ort: 26.03.2026, Maintal

Unterschrift: _____